

Introduction to Modern Causal Inference

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Introduction

Goals and Approach

Our goal is to get people with any amount of experience in formal mathematics past undergraduate probability to be able to take a vaguely-defined scientific question, translate it into a formal statistical problem, and solve it by constructing a tool that is optimal for the job at hand.

There are many texts and courses that share the same philosophy and have substantial overlap in terms of the pedagogic approach and topics. In particular, we owe a lot to the biostatistics curriculum at UC Berkeley and to more encyclopedic texts like [*Targeted Learning*](#). Nonetheless, we think that the particular selection of topics and tone in this book will make the content accessible enough to new audiences that it's worthwhile to recombine the material in this way.

Philosophy

This book is rooted in the philosophy of modern causal inference. What sets this philosophy apart are the three following tenets:

1. The first is that for most practical purposes, **the point of statistics *is* causal inference**. Ultimately, we humans are concerned with how to make decisions under uncertainty that lead to the best outcome. These are fundamentally causal questions that ask “*what if*” we did A instead of B? The machinery of statistical inference is agnostic to causality, but that doesn't change the fact that our motivation in using it isn't. Therefore there is little point in shying away from causal claims because noncausal claims are not often of any practical utility.
2. That leads to the second, often misunderstood point, which is that **there is no such thing as a method for causal inference**. This isn't a failing of the statistics literature so much as a failing of the naive popularization of new ideas (e.g. blog posts, low-quality papers). The process of statistical inference (for point estimation) only cares about estimating a parameter of a probability distribution and quantifying the sampling distribution of that estimate. There is nothing “causal” about that, and many are surprised to learn that the algorithms used for “causal” inference are actually identical to those used to make noncausal statements. What makes an analysis causal has little if anything

to do with the estimator or algorithm used. It has everything to do with whether or not the researcher can satisfy certain non-testable assumptions about the process that generated the observed data. It's important to keep these things (statistical inference and causal identification) separated in your mind even though they must always work together.

3. The last and perhaps greatest shortcoming of traditional statistics pedagogy is that it generally does not teach you to ask the question that makes sense scientifically and *then* translate it into a statistical formalism. This is partly because, in the past, our analytic tools were quite limited and no progress could be made unless one imposed unrealistic assumptions or changed the question to fit the existing methods. Today, however, **we have powerful tools that can then translate a given statistical problem into an optimal method for estimation** in a wide variety of settings. It's not always totally automatic, but at a minimum it provides a clear way to think about what's better and what's worse. It liberates you from the route if-this-kind-of-data-then-that-method thinking that only makes for bad statistics and bad science.

Taken together, these three points give this movement a clear, unified, and increasingly popular perspective on causal inference. We have no doubt whatsoever that this is the paradigm that will come to dominate common practice in the next decades and century.

Pedagogy

The modern approach to causal inference is already well-established and discussed at length in many high-quality resources and courses. So why have yet another book going over the same material?

Well, there are many different kinds of students with different backgrounds for whom different presentation styles work better! We've chosen an approach for this particular text that focuses on the following principles:

1. **Rigor with fewer prerequisites.** Many students are turned off or intimidated by the amount of mathematical formality that is required to rigorously understand modern causal inference. While you don't actually need a lot of math to use and understand the ideas, you do need *some* math to be able to work independently in the field. Since one of our goals is for you to be able to *construct your own* efficient estimator for a never-before-seen problem, we have to think precisely and not hand-wave away rigor.

For better or worse, the math you need is built on several layers of prerequisites- real analysis, probability theory, functional analysis, and asymptotic statistics (more on that later). In an ideal world, we would all have taken these courses and be prepared for what's next, but in practice we know many people have gaps in their knowledge.

In this book we try to address that by providing a lot of foundational background information about these topics as they come up. We do our best to build up intuition for the

component pieces of things before diving into implications. We lean heavily on figures and natural english prose to explain these concepts alongside the formal mathematical notation. We emphasize intuition and practical use rather than formalism. At a minimum, we try to be very clear about what the foundations actually *are* so that if you don't understand something you at least know where to look.

2. **Core concepts.** Modern causal inference is a huge field. There are so many exciting applications and theoretical developments that it can be hard for students to find their way at first.

We've tried to keep thing as short and concise as possible in this book by focusing on core ideas and a limited number of examples. The point of this is to emphasize what is most important and to allow you to learn the fundamentals without getting distracted. That means there are many topics that aren't covered in this book (see section below), but the idea is that this book will give you the tools and mindset you need in order to continue your learning productively.

3. **A welcoming voice.** Modern causal inference is fast-moving and intense. It's easy to feel like you don't belong or aren't good enough to participate. The agnostic, impersonal tone that is best for writing a clear academic paper doesn't always create the warm, emotional connection that is necessary for us to feel secure and ready to learn.

To deal with that problem, the voice we use throughout this book is informal and decidedly nonacademic. We write in the first person and address you, the reader, directly. Figures are hand-drawn and cartoonish. We provide personal commentary alongside agnostic technical material. All of this is deliberate and meant to help you feel comfortable and guided through this world. You are not alone in learning!

If you find this book isn't for you, that's also great! Here is a list of other resources that cover some of the same ground:

- [*Semiparametric Doubly Robust Targeted Double Machine Learning: A Review* - Edward Kennedy](#)
- [*Semiparametric Theory and Missing Data* - Anastasios Tsiatis](#)
- [*Targeted Learning* - Mark van der Laan, Sherri Rose](#)
- [*Unified Methods for Censored Longitudinal Data and Causality* - Mark van der Laan, James Robins](#)
- [*Causal Inference: What If?* - Miguel Hernán, James Robins](#)

Rigor with Fewer Prerequisites

Despite our best efforts, you will need *some* background material to get the most out of this book. The “hard” prerequisites listed are non-negotiable (this book will make no sense to you at all if you don't know what a probability distribution is), but the “soft” prerequisites are

completely optional. We don't cover these topics here because there are already so many good resources to learn them and there's no need to reinvent the wheel. We provide the curated references below so you know where to go to fill in the holes in your understanding as you go along.

Hard Prerequisites You'll need to know probability theory at the undergraduate level. Random variables, expectation, conditional expectation, probability densities should be familiar to you, and ideally you have at least a vague understanding of what convergence in distribution means.

You also need undergraduate multivariable calculus and linear algebra: integrals, derivatives, vectors, etc. should all be well-worn tools at your disposal.

If you're shaky on these topics, we recommend [Khan Academy](#). We also encourage you to check out the real analysis and measure-theoretic probability resources listed below, which will provide you with a very useful conceptual understanding.

Soft Prerequisites The following topics aren't necessary to grasp the arc of the arguments we'll make, but a fully rigorous understanding isn't possible without them. As we go along, we'll provide parenthetical commentary to elucidate concepts from these subjects that you may not be familiar with. We suggest attempting to read and understand this book all the way through and taking notes on the places where you feel like you're missing something. This will help you prioritize your time if you choose to fill in the gaps. None of this material is beyond you, I promise. It just takes a little time to work through it. I learned it all from these books myself: you may be surprised to hear I don't have any formal training in mathematical statistics!

These books and videos give what I think is the most self-contained treatment of their respective topics that is in line with the didactic approach taken in this book. They tend to have a more conversational tone, provide more background and intuition to the reader, and have plenty of worked examples. As always, however, there are many alternatives that may work better for you depending on your needs. These subjects should be tackled in the order they are listed here:

Field	Topics to Focus On	Resources
1. Intro Real Analysis	sequences, limits, series, continuity, convergence (pointwise, uniform), derivative	Understanding Analysis (Ch 1-6) - Abbott

Field	Topics to Focus On	Resources
2. Measure-Theoretic Probability	definition and construction of measure, Lebesgue integral and its properties, Radon-Nikodym theorem, change-of-variables formula, modes of convergence for random variables, independence, central limit theorem, conditional expectation	Measure Theory Youtube Lectures - Landim, Measure, Integral, and Probability - Capiński & Kopp
3. Functional Analysis in $\mathcal{L}_2$	projection, bounded dual space, Reisz representation	Introductory Functional Analysis with Applications (Ch 1-3) - Kreyszig

A few tips for self study: I recommend reading skimming each chapter first and getting an idea of what the main ideas are and how they interconnect to each other and to what you've already done. Take your time to understand what the definitions of the objects are that are used in each theorem (e.g. what exactly is a sequence / random variable / linear bounded functional?). Think of some examples and draw pictures.

Then go through the chapter again and read it slowly, line by line. Reading a line of math should take you as long as it takes to read a paragraph of english because that's often how much information is conveyed therein. Now draw some pictures or think of examples attempting the illustrate the content of the theorems. After you're done, do several of the exercises that the author provides. I've found that it also helps to make a "one-page summary" or "cheatsheet" of all the material in the chapter (definitions, important theorems, and their relationships). This will help you keep all the concepts organized as you go along.

I'll also refer to [Asymptotic Statistics](#) by A.W. van der Vaart in many places throughout the guide where we need to point to technical material. Relative to the way we write things up here, that book goes over background material much faster, so you may find it more challenging to read through. However, it does contain all the necessary information and details. If you've read the material in the table above and you really want to dot your i's and cross your t's on the material in this book you can try your hand at ch. 1, 2, 6, 7, 8, and 25 of Asymptotic Statistics.

Core Concepts

We've done our best to keep this book lean and targeted on foundational concepts. As a consequence, there are many important and interesting topics that we'll say nothing or little

about. For the sake of completeness, we'll list some of them here so you know where to start looking when you're ready to build on your solid foundation.

To keep things simple and consistent, most of the examples in this book focus on inference of the average treatment effect in simple covariate-treatment-outcome observational or randomized studies. However, there are many data structures which are much more elaborate than this. For example, we do not fully discuss inference in longitudinal settings with time-varying covariates and treatments, inference when some outcomes are censored, stochastic interventions, or continuous treatments. Non-IID data (e.g. treatment spillover in a network) is another interesting topic that is not covered. Nonetheless, what is in this book will prepare you to study all of these topics and to understand them as special cases or extensions of the general approach presented here.

While the tools we describe in this book are extremely powerful, they are not *all*-powerful. Some estimands of legitimate interest are not “pathwise differentiable” and therefore cannot be analyzed with these tools. An important example of such an estimand is the conditional average treatment effect, which is the subject of extensive research. There is not yet a unified approach to these problems.

Another interesting and useful setting that we do not discuss is online (or reinforcement) learning. Many topics related to this literature (e.g. off-policy evaluation) are directly addressable with the methods in this book, but here we omit all discussion of scenarios where the treatment policy can be manipulated during data collection as more information comes in. The framework for causal inference we develop in this book does extend to such settings but only with some added complexity that is best understood after learning the basics.

Lastly, the reader should be aware that this book discusses the most popular and widely used framework for causal inference, which is that of potential outcomes in the frequentist population model. Readers may have heard of directed acyclic graphs (DAGs) or nonparametric structural equations models (np-SEM) as alternatives to potential outcomes, but these frameworks are in fact largely equivalent for the vast majority of practical problems so it is really not worth worrying about the differences until you are very deep in the weeds.

Potential outcomes are also used outside the population model in the randomization inference framework, which differs in that only the treatment assignment is assumed to be random (i.e. sampling from a larger population is not a source of variability). Much of what you learn here is also applicable to randomization inference. Potential outcomes are also commonly used in Bayesian approaches to causal inference, which we also don't discuss here.

Topics

This book aims to be a relatively self-contained treatment of the modern approach to causal inference. As we said above, the goal is for you to be able to take a vaguely-defined scientific question, translate it into a formal statistical problem, and solve it using a tool that you

optimally construct for the job at hand. Thankfully, there's a well-developed process for doing that called the [Statistical Roadmap](#) which we follow closely in the chapters of this book:

1. To begin with, we need to first understand the process of *statistical inference*. We do that by defining a set of assumptions that represent what we know is true about the real world along with some true quantity of interest. Only then do we propose some kind of method or algorithm that, given data, produces an estimate of this quantity. We should be able to show that, with enough data, our answer gets closer and closer to the truth and that we have some way to quantify the uncertainty due to random sampling. These are the topics we'll work through in chapter 1, closing out with an example that illustrates the danger of blindly applying popular methods without thinking through the question and assumptions.
2. Once we've understood the goal and process of inference, we introduce causality. Causal inference questions are just statistical inference questions but in an imaginary world where we can always observe the result of alternative "what if"s. Of course in the real world we can't see all the "what if"s. We only get to see what happened, not what *could* have happened had we made a different choice, taken a different drug, or implemented a different policy. To link the two worlds together we have to make some assumptions to guarantee that the answer to a statistical inference problem in the real world actually represents the answer to our causal question in the world of "what if"s. Chapter 2 lays out this process (*causal identification*) and gives examples. Chapter 2 also addresses why a well-defined intervention is important to formulating something as a causal question and discusses broadly applicable methods that quantify the impact of potentially erroneous identifying assumptions.
3. Once we have an identification result in hand, we are completely done with causality and can focus on solving the real-world inference problem. But for better or for worse there are thousands of inference methods, each of which work under different assumptions and target different parameters. Choosing among these is daunting and often puts the cart before the horse because we're forced to define our question such that an existing method can solve it. Chapter 3 lays the foundation to address this problem by showing that for a wide range of assumptions and questions, there is generally a way to quantify what the best possible estimator is in terms of the *efficient influence function*. The efficient influence function completely describes the behavior of the optimal estimator in large samples. This optimal estimator always converges to the right answer with more data and has the lowest legitimate uncertainty possible under the given assumptions. We work through examples of deriving efficient influence functions for the average treatment effect in standard observational and randomized experiments.
4. Finally, chapter 4 does the hard work of putting the theory from chapter 3 into practice. Here we show how we can actually build an *efficient estimator*- one which has the efficient influence function. In other words, given a statistical problem, we can construct an optimal method to solve it. We outline the three main strategies for doing this, which are called bias-correction, estimating equations, and targeted maximum likelihood. We show the different ways in which these strategies ensure that the constructed estimator

has the right behavior. All of them ultimately attain the same goal and are theoretically equivalent in large-enough samples, but the differences are relevant for how the constructed estimators perform in practice.

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Part I

Efficiency Theory

So far we’ve learned how to pose a good causal question and we’ve examined what kinds of causal assumptions we might have to make in order to translate the causal estimand to an observable estimand. Now that we have an observable estimand, we’re done with causality for the time being (until we return to interpreting our result in context of the causal assumptions). Now we are left with a purely statistical problem: we want to estimate some aspect of a distribution given some finite samples from that distribution.

The fact that identification and estimation can be cleanly separated is a wonderful feature of the modern approach to causal inference. The causal problem is that we don’t observe the right “columns” (the potential outcomes) in the “population dataset”. The statistical problem is that we don’t observe all of the “rows”.

[footnote] cite the uniform convergence thing in RCTs as a counterexample where ID and estimation interplay

In this chapter we will examine how to define whether an estimator is in some sense “optimal” for a given estimand. If you’ve already done some applied causal inference, you probably already know that there are many different practical approaches for estimating causal effects (matching, regression, inverse probability weighting, etc.). How can we judge these against each other when in reality we never observe the “truth” that all of them are trying to estimate? What reason might we have to choose one over another? Once we’ve developed a characterization of optimality, we will see in the next chapter how to use that knowledge to construct optimal estimators for many kinds of estimands.

As a heads up, this chapter is technical, so you will see a lot of pointers to the appendices, which explain foundational concepts in mathematical statistics. As a reminder, these are targeted at readers who have taken a course in basic probability and statistics, but perhaps not real analysis or measure-theoretic probability. The appendices by no means cover all of what there is to know about their respective topics. While I strive to be as rigorous as possible, the primary goal is not rigor but instead to give the reader a sense of why and how the concepts are most often used in asymptotic statistics so that they can begin to distinguish between the main ideas and the mathematical details in the literature.

1 Asymptotic Linearity

1.1 What do we want from estimators?

Let's forget about what we know about estimators for a second and start from scratch. What is it that we *want* from an estimator in the first place? Can we set some “dealbreaker” conditions and some “nice-to-haves”?

Ultimately any estimator $\hat{\Psi} : \mathbb{P}_n \rightarrow \mathbb{R}$ is characterized entirely by its sampling distribution at a given n , for a given P i.e. the distribution of the random variable $\Psi(\mathbb{P}_n)$. What would we want these sampling distributions to look like relative to some truth $\Psi(\mathbb{P})$? I invite you to stop reading at this point and brainstorm some ideas.

To start with, let's shoot the moon: why not demand that the estimate is always *exactly* equal to the estimand, i.e. $\hat{\Psi}(\mathbb{P}_n) = \Psi(\mathbb{P})$ for all distributions $\mathbb{P}$ in the model $\mathcal{P}$? That might seem completely absurd to you but it's worth taking a second to ask why that is really such a crazy thing to demand. It is less extreme to ask that our estimate is correct *on average*, but even this is usually not possible to do. So let's start with asking that our estimator is *asymptotically* unbiased: the estimate is more or less right on average given a large-enough sample size. Informally: $\mathbb{E}[\hat{\psi}] \approx \psi$ for n large enough.

If I have two estimators that are both unbiased, which is better? Since the total estimation error $\mathbb{E}[(\hat{\psi} - \psi)^2]$ can be decomposed to $(\mathbb{E}[\hat{\psi}] - \psi)^2 + \mathbb{V}[\hat{\psi}]$ (by adding and subtracting $\mathbb{E}[\hat{\psi}]$ inside the square and expanding), the unbiased estimator with the smallest overall error will be the one with the smallest variance. Therefore we should desire estimators that have small (asymptotic) sampling variance, subject to being (asymptotically) unbiased.

Let's throw in a practical consideration. When we estimate a quantity we usually report confidence intervals or p-values alongside the point estimate. Those inferential statistics depend on having some estimate of the sampling variance of our estimator, which we usually construct based on a theoretical understanding of what that sampling distribution should be. Many commonly used estimators are known to be asymptotically normal, which greatly facilitates inference. Therefore another desired criterion might be that our estimator should have a theoretically understood sampling distribution.

Let's summarize our list of what we want out of our estimator so we can easily refer back:

1. asymptotically unbiased
2. small asymptotic sampling variance

3. well-understood sampling distribution

1.2 Asymptotic Linearity

Given these criteria, let's simplify our problem of finding an optimal estimator to finding an optimal estimator among those that are *asymptotically linear*. I'll lay out what that means, argue for why it's a reasonable restriction given our criteria above, and then explain why this restriction is helpful in characterizing optimality.

Asymptotic Linearity

Definition 1.1. As before, assume that Z_i are drawn IID from some distribution $\mathbb{P}$. An *asymptotically linear* estimator $\hat{\psi}$ is one that can be written like

$$\hat{\psi} = \psi + \frac{1}{n} \sum_i \phi(Z_i) + o_{\mathbb{P}}(n^{-1/2})$$

for some mean-zero function ϕ with mean zero $\mathbb{E}[\phi(Z)] = 0$ and finite variance $\mathbb{V}[\phi(Z)] < \infty$.

Warning

If you don't really understand what $o_{\mathbb{P}}(n^{-1/2})$ means in the definition above or why it implies the sample mean term “dominates” asymptotically, you should work through all of [Appendix A](#) before returning here.

The function ϕ is called the *influence function* of the estimator $\hat{\psi}$. A better name for it could be the “averaging transformation” of the data for a given estimator because once the data have been transformed by ϕ , the estimator can be closely approximated by the truth plus a sample average of those transformed variables (which have mean zero). In large samples, sample averages of mean-zero variables behave like Gaussian random noise. It's somewhat remarkable that a huge number of very complicated estimators actually behave this way, as we will see shortly. By definition, every asymptotically linear estimator has an influence function that satisfies the above criteria: having one is what makes the estimator asymptotically linear.

Example 1.1 (Sample Mean). The sample mean estimator is one example of an asymptotically linear estimator. To see that, write $\hat{\psi} = \frac{1}{n} \sum Z_i$ as

$$\hat{\psi} - \psi = \frac{1}{n} \sum \underbrace{(Z_i - \psi)}_{\phi(Z_i)} + 0.$$

For this estimator, the remainder term is exactly 0, which clearly satisfies the requirement of being $o_{\mathbb{P}}(n^{-1/2})$.

The influence function for an estimator also depends on the distribution of the data. For example, in the case of the sample mean we saw that $\phi(z) = z - \psi$ which we should really write as $\phi_{\mathbb{P}}(z) = z - \Psi(\mathbb{P})$ to emphasize that this particular influence function depends on the mean of Z under the distribution $\mathbb{P}$. Often we write something like $\phi_{\mathbb{P}}(z)$ or even $\phi(\mathbb{P})(z)$ to mean the influence function of a given estimator *at a particular distribution* $\mathbb{P}$. But more often than not we leave the $\mathbb{P}$ part implicit to minimize notational clutter.

In fact, it's more appropriate to say that an estimator is asymptotically linear *for some set of distributions*, i.e. within some model. But usually the restriction on the set of distributions is not particularly meaningful so we just say the estimator is “asymptotically linear (under regularity conditions)” and leave it at that.

Exercise 1.1. Let's presume that Z are drawn from a distribution $\mathbb{P}$ such that $\sqrt{n}(\frac{1}{n} \sum Z_i)$ diverges as a random variable (see Section A.2.2 for a definition of divergence of random variables). A “textbook example” of such a distribution is Pareto(1, 1.5), but this happens for many “fat-tailed” distributions. You can ask your favorite large language model if you want to see a proof.

Why does this tell you that the sample mean cannot be asymptotically linear at $\mathbb{P}$?

The point of this example isn't to convince you that the sample mean isn't asymptotically linear (it is for essentially all distributions we care about), it's just to cement the idea that asymptotic linearity of an estimator technically depends on what distributions you're talking about and to better understand the definition by considering a counterexample.

It's totally possible for two different estimators to have the same influence function (at each distribution). This is because two different algorithms can turn out to behave very similarly as the sample size increases. For example, we could construct a very silly alternative estimator for the mean by adding $1/n$ to our sample mean estimate. This extra term is certainly converging to zero at better than a $\sqrt{n}$ rate so it is negligible and the influence function for our alternative estimator is the same as for the sample mean. We'll also see the idea of two different estimators sharing an influence function come up later when we discuss different strategies for building optimal estimators: the estimators we get from using each strategy may be different, but they all have the same influence function and therefore all behave similarly in large-enough samples.

At this point you may be worried about how you can take your favorite estimator and derive its influence function. It's often possible to prove an estimator is asymptotically linear, but in this book we actually won't need to do that. We're going to go the other direction: starting from the estimand, we're going to come up with a way to determine the “best” possible influence function and then *construct* an estimator around it.

1.3 Benefits of AL Estimators

Given our desiderata, asymptotically linear (AL) estimators are a very sensible class of estimators to restrict ourselves to. For one, AL estimators are guaranteed to be consistent: that is, they converge to the estimand in probability and are thus asymptotically unbiased. This is guaranteed by the fact that any influence function ϕ is mean-zero so the associated sample mean converges to 0 by the law of large numbers.

Exercise 1.2. Consider the silly estimator $\hat{\psi} = 0$ when attempting to estimate the population mean $\mathbb{E}[Z]$ in a model containing all possible finite-mean distributions of Z . Is this estimator asymptotically linear across the entire model? Why or why not?

Another related benefit of asymptotically linear estimators is that they satisfy our practical requirement of understanding the sampling distribution. By the central limit theorem, $\psi + \mathbb{P}_n \phi$ looks more and more like $\psi + \mathcal{N}(0, \mathbb{V}[\phi(Z)]/n)$ as the sample size increases. The remaining term $o_{\mathbb{P}}(n^{-1/2})$ is negligible in comparison (by definition). More often you will see this written formally as

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathcal{N}(0, \sigma_*^2)$$

where $\sigma_*^2 = \mathbb{V}[\phi(Z)] = \mathbb{P}\phi^2$ (since ϕ has mean zero).

We call σ_*^2 the *asymptotic variance* of our estimator. You should think of $\hat{\psi} - \psi$ as a normal distribution that is shrinking towards a spike at 0: its variance is decreasing at a rate of $n^{-1/2}$ by the central limit theorem (see Example A.4 in Section A.2.2). We therefore “stabilize” the shrinking distribution of $\hat{\psi} - \psi$ by “blowing it up” by a factor of $\sqrt{n}$. The asymptotic variance σ_*^2 is the variance of this stable normal distribution (compare rows 1 and 3 of the figure in Figure A.2). It is *not* the limit of the sampling variance $\mathbb{V}[\hat{\psi}]$ as a function of n because that limit is always zero on account of the fact that the estimate is converging to the estimand.

At any finite sample, the standard error of the estimate is roughly given by $\sigma = \frac{\sigma_*}{\sqrt{n}}$. Since the asymptotic variance is given by the variance of the influence function, understanding the influence function is all we need to understand the sampling distribution.

Constructing asymptotically valid confidence intervals and p-values for any asymptotically linear estimator is therefore as simple as estimating $\sigma_*^2 = \mathbb{P}\phi^2$. If we can construct an estimate $\hat{\sigma}_*$ that is close to σ_* for large-enough n then we can estimate the standard error of our estimate with $\hat{\sigma} = \frac{\hat{\sigma}_*}{\sqrt{n}}$ we will get asymptotic 95% coverage of ψ with an interval of the form $[\hat{\psi} - 2\hat{\sigma}/\sqrt{n}, \hat{\psi} + 2\hat{\sigma}/\sqrt{n}]$ (this is a “Wald-type” confidence interval).

Exercise 1.3 (Wald Interval).

- a. Assuming that $\sqrt{n}(\hat{\psi} - \psi) \sim \mathcal{N}(0, \sigma^2)$, prove that $\hat{\mathbf{C}} = [\hat{\psi} \pm 2\sigma/\sqrt{n}]$ is a 95% confidence interval, i.e. $\mathbb{P}\{\psi \in \hat{\mathbf{C}}\} > 0.95$.
- b. Assuming asymptotic linearity, how does our actual knowledge of the sampling

- distribution of $\hat{\psi}$ compare to what we assumed in part (a)?
- c. In the real world our confidence interval is $[\hat{\psi} \pm 2\hat{\sigma}/\sqrt{n}]$. How does this compare to what we used in part (a)?
 - d. Given (b) and (c), do you think the real-world confidence interval will generally have better-than 95% coverage for all sample sizes?

Example 1.2 (Variance of the sample mean). In the case of the sample mean estimator $\hat{\psi} = \frac{1}{n} \sum Z_i$ we previously saw the influence function was $\phi(z) = z - \psi$. A plug-in estimate of $\mathbb{P}\phi^2$ is therefore given by $\frac{1}{n} \sum \hat{\phi}(Z_i)^2 = \frac{1}{n} \sum (Z_i - \hat{\phi})^2$

Example 1.3 (Variance of ordinary least squares coefficients). Let's say we're trying to estimate the regression coefficients in a linear model. Let $(Y, X) \sim \mathbb{P}$ with $X \in \mathbb{R}^d$. Let $\beta_0 = \operatorname{argmin}_{\beta} \mathbb{E}[(Y - X\beta)^2]$ be the coefficients that minimize the population mean squared error. This will be our estimand. We haven't assumed the Y vs. X relationship is truly linear, we're just estimating the best linear approximation. As an estimator we will use the ordinary least squares (OLS) estimate $\hat{\beta} = \operatorname{argmin}_{\beta} \frac{1}{n} \sum [(Y_i - X_i\beta)^2]$.

It is a well-known result that the OLS estimator is asymptotically linear under mild regularity conditions. For one proof of this see chapter 5 of Asymptotic Statistics (but it's not important at the moment). The influence function is

$$\phi(x, y) = \mathbb{E}[XX^\top]^{-1}x(y - x^\top\beta)$$

You may recognize this because it's where the “sandwich” estimator (also known as “Huber-White” or “robust” estimator) of the standard error comes from. The (asymptotic) sandwich standard error is a plug-in estimate for the quantity $\mathbb{E}[XX^\top]^{-1}\mathbb{E}[X(Y - X^\top\beta)^2X^\top]\mathbb{E}[XX^\top]^{-1}$, which is nothing other than $\mathbb{P}\phi\phi^\top$, i.e. the equivalent of $\mathbb{P}\phi^2$ when ϕ is a vector-valued function (as is the case when the estimand is a vector).

In this example the dependence on $\mathbb{P}$ is through the expectation $\mathbb{E}[XX^\top] = \mathbb{E}_{\mathbb{P}}[XX^\top]$ and the true best regression coefficients $\beta = \beta_{\mathbb{P}}$. As usual, I've left off the explicit dependence on $\mathbb{P}$ in the notation.

The interesting thing about this example is that computing the estimate involves a fairly complex algorithm: either gradient descent or a series of matrix multiplications and inversions. It's not a “simple” estimator like the sample mean. Nonetheless, in large enough samples, the estimator behaves as if it were just a sample mean of a transformation of the data.

OLS is a special case of a huge class of estimators called M-estimators. Others include sample quantiles, robust regressions, and many others. All of these are asymptotically linear under mild regularity conditions.

In our list of desired characteristics we also wrote that we wanted estimators to have as small a variance as possible. Asymptotic linearity by itself doesn't guarantee small variance, but it does give us a very easy way to compare two estimators. If we have estimators $\hat{\psi}_1$ and $\hat{\psi}_2$ which are both asymptotically linear, then all we need to know to decide which one is better (in large-enough samples) is to compare their asymptotic variances which are given by $\mathbb{P}\phi_1^2$ and $\mathbb{P}\phi_2^2$.

You may also recognize $\mathbb{P}\phi^2$ as the squared $\mathcal{L}_2(\mathbb{P})$ norm of ϕ . Therefore we are comparing $\|\phi_1\|$ to $\|\phi_2\|$. In other words, the better estimator is whichever has the “smaller” influence function! In the following sections this is the key insight we will use to construct optimal estimators.

Warning

If you are unfamiliar with $\mathcal{L}_2$ spaces or don't understand why $\mathbb{P}\phi^2$ is a norm, you should work through Section [B.1](#) and Section [B.2](#) now.

The theory of optimality we are building is only valid for asymptotically linear estimators. If there were many practically useful estimators that were *not* AL, this could be a problem because we would not be able to usefully compare them. We'll discuss this more in a later section but the short answer is that AL estimators are an extremely broad class and we're not leaving much of anything on the table.

2 Regularity

While restricting ourselves to asymptotically linear estimators is a good start, unfortunately it is not quite enough to develop a sensible theory of optimality. In this section we'll first argue for why a further restriction is not only sensible but useful. Then we will come up with and refine a definition that suits our needs.

2.1 Hodges' Estimator

Let's presume we're interested in estimating the mean $\psi = \Psi(\mathbb{P}) = \mathbb{E}_{\mathbb{P}}[Z]$ from n IID draws of a scalar random variable Z that we know is normally distributed with standard deviation $\sigma = 1$. Our statistical model is therefore $\mathcal{P} = \{\mathcal{N}(\psi, 1) : \psi \in \mathbb{R}\}$, a one-dimensional parametric model. Obviously this is a contrived example where we know a lot about Z , but the point is to show that even in this extremely favorable setting we can be led astray by mathematical arguments that put our attention in the wrong place.

We'll consider two estimators:

$$\begin{aligned}\hat{\psi} &= \frac{1}{n} \sum Z_i \\ \tilde{\psi} &= \begin{cases} \hat{\psi} & |\hat{\psi}| > n^{-1/4} \\ 0 & |\hat{\psi}| \leq n^{-1/4} \end{cases}\end{aligned}$$

The first estimator is just the usual sample mean. The second is what is called *Hodges' estimator*. It is an alternative that usually behaves similarly to the sample mean in large samples, but prefers to estimate 0 in small samples.

[pic: hodges as a fn of sample mean for different n]

Since Z are normal random variables with unit variance and mean ψ , we know that the sample mean has the exact distribution $\mathcal{N}(\psi, 1/n)$. Thus it follows that

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathcal{N}(0, 1)$$

Hodges' estimator, however, has the following asymptotic distribution:

$$\sqrt{n}(\tilde{\psi} - \psi) \rightsquigarrow \begin{cases} \mathcal{N}(0, 1) & \psi \neq 0 \\ \mathcal{N}(0, 0) & \psi = 0 \end{cases}$$

Derivation of the asymptotic distribution of Hodges' estimator

Let's prove that $\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathcal{N}(0, 0)$ when the mean of Z is zero. This will require some understanding of what is meant formally by convergence in probability and a common probability tool called Chebyshev's inequality. It's not at all important to understand this proof, I just provide it for completeness and as a refresher for those who are familiar with this kind of argument.

That statement is the same as saying that $\sqrt{n}(\hat{\psi} - \psi) \xrightarrow{\mathbb{P}} 0$, which we will prove. For any $\epsilon > 0$,

$$\begin{aligned} \mathbb{P} \{ |\sqrt{n}\tilde{\psi}| > \epsilon \} &= \mathbb{P} \{ 1(|\hat{\psi}| > n^{-1/4}) |\hat{\psi}| > \epsilon/\sqrt{n} \} \\ &= \mathbb{P} \left\{ |\hat{\psi}| > \epsilon/\sqrt{n} \mid |\hat{\psi}| > n^{-1/4} \right\} \mathbb{P} \{ |\hat{\psi}| > n^{-1/4} \} \\ &\leq \mathbb{P} \{ |\hat{\psi}| > n^{-1/4} \} \end{aligned}$$

by definitions and total probability. Now, applying Chebyshev's inequality,

$$\mathbb{P} \{ |\hat{\psi}| > n^{-1/4} \} \leq \frac{\sigma^2/n}{n^{-1/2}} = \frac{1}{\sqrt{n}} \rightarrow 0$$

Which proves the desired convergence. Perhaps surprisingly, the proof still works if we are interested in proving that $r_n(\hat{\psi} - \psi)$ converges in probability to 0 when the mean of Z is zero for an arbitrary rate r_n . In other words, this convergence happens arbitrarily quickly as far as the asymptotics are concerned.

Nearly identical arguments are used to show that Hodges' estimator has the same asymptotic distribution as the sampling mean when the true mean is not zero. This is a fine exercise to try on your own to practice your probability theory.

Like the sample mean, Hodges' estimator is definitely asymptotically linear: the influence function is

$$\phi_{\mathbb{P}}(z) = \begin{cases} z - \mathbb{E}[Z] & \mathbb{E}[Z] \neq 0 \\ 0 & \mathbb{E}[Z] = 0 \end{cases}$$

which is certainly a $\mathbb{P}$ -mean-zero $\mathcal{L}_2(\mathbb{P})$ function for all $\mathbb{P}$ with finite mean of Z . Therefore if we were defining optimality among AL estimators, we would have to consider Hodges' estimator as a candidate.

Exercise 2.1. Judging by the asymptotic distributions, do you think the sample mean or Hodges' estimator is better?

The sample mean and Hodges' estimator are asymptotically the same across most distributions. But in any case where $\mathbb{E}[Z] = 0$, Hodges' is clearly better: its variance goes to zero even if

blown up by the square root of n whereas the sample mean's variance does not. Therefore by these arguments Hodges' is the better estimator overall.

But if this is true, why do we use the sample mean instead of Hodges' estimator? And moreover why couldn't we reason the same way about a Hodges'-style estimator that has a special preference for some other value instead of zero? Wouldn't that estimator also be better than the sample mean for that special value and otherwise the same?

The reason we don't use Hodges' estimator is that the asymptotic arguments hide that this estimator actually performs much worse than the sample mean for most values of n and most possible distributions of Z . You can see this in a simulation (or work it out analytically):

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
hodges = function(Z) {
  n = length(Z)
  ifelse(abs(mean(Z)) > n^(-1/4), mean(Z), 0)
}
```

```
sim = function(n, psi) {
  Z = rnorm(n, mean=psi)
  tibble(
    n = n,
    psi = psi,
    mean = mean(Z),
    hodges = hodges(Z)
  )
}
```

```
params = expand_grid(
  n = c(10, 100, 1000),
  psi = seq(-1, 1, 0.01),
```

```

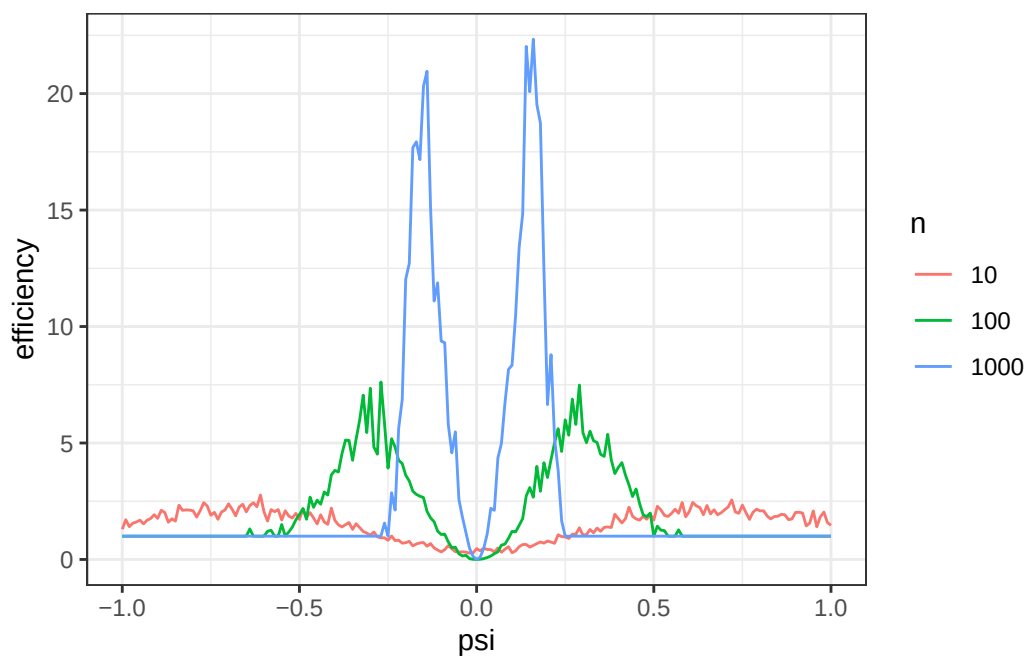
)

result = 1:100 |>
  map_df(~pmap_df(params, sim))

result |>
  group_by(n = factor(n), psi) |>
  summarize(
    mse_mean = mean((mean - psi)^2),
    mse_hodges = mean((hodges - psi)^2),
    efficiency = mse_hodges / mse_mean
  ) |>
ggplot() +
  geom_line(aes(psi, efficiency, color=n)) +
  theme_bw()

```

`summarise()` has grouped output by 'n'. You can override using the `groups` argument.



The y -axis of the plot shows the relative efficiency of Hodges' estimator, defined as the ratio of MSEs between Hodges' estimator and the sample mean: $\mathbb{E}[(\tilde{\psi} - \psi)^2] / \mathbb{E}[(\hat{\psi} - \psi)^2]$. If this is

greater than one, then Hodges' estimator is better than the sample mean, if less than one, it is worse.

As you can see, regardless of what n is, Hodges' estimator is worse (and sometimes *much* worse) than the sample mean for the majority of the plotted values of ψ . So then why do the asymptotic arguments suggest that Hodges' estimator is actually better than the sample mean?

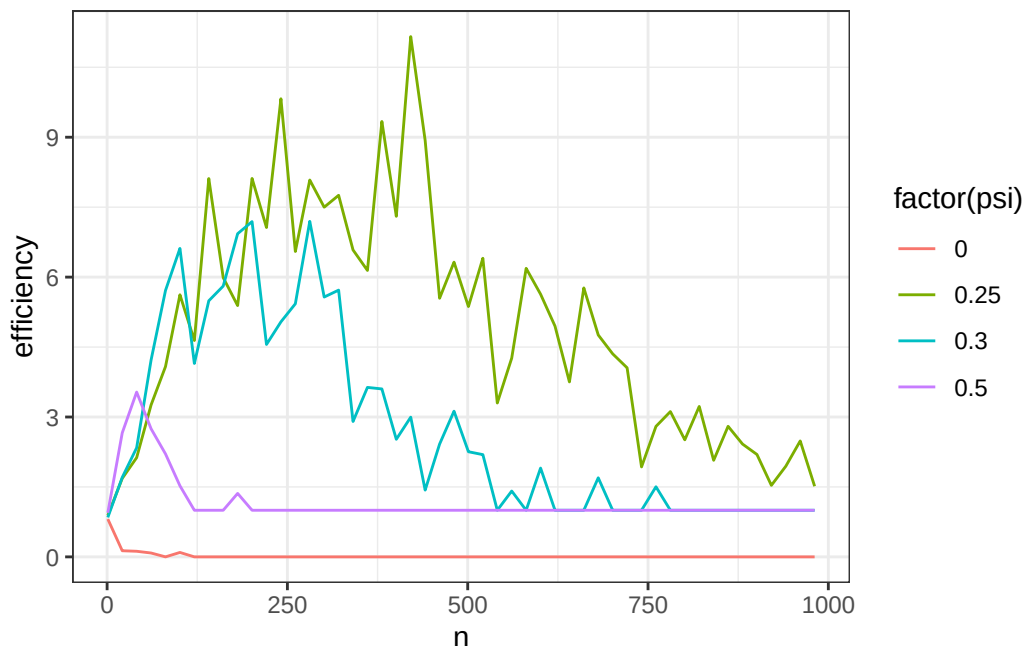
The problem is that the asymptotics are only concerned with the behavior of each estimator at any particular distribution of Z . In our contrived model, the distributions of Z are all normal with unit variance so a single distribution corresponds to a value of ψ . If you look at a single nonzero point on the x -axis (say, $\psi = 1/2$) and compare the relative efficiency as n increases you'll see that it does converge to 1: both estimators are asymptotically equivalent. At the point $\psi = 0$, however, the relative efficiency converges to 0 meaning that Hodge's estimator is infinitely more efficient at that point asymptotically. We can repeat our simulations but plot them a little differently in order to see this. Instead of plotting efficiency against ψ and slicing by n , we'll instead plot efficiency against n and slice by ψ (you could imagine a 3D plot with efficiency on the z-axis and n and ψ on the x and y).

```
params = expand_grid(
  n = seq(1, 1000, by=20),
  psi = c(0, 0.25, 0.3, 0.5)
)

result = 1:100 |>
  map_df(~pmap_df(params, sim))

result |>
  group_by(n = n, psi = psi) |>
  summarize(
    mse_mean = mean((mean - psi)^2),
    mse_hodges = mean((hodges - psi)^2),
    efficiency = mse_hodges / mse_mean
  ) |>
ggplot() +
  geom_line(aes(n, efficiency, color=factor(psi))) +
  theme_bw()
```

``summarise()`` has grouped output by 'n'. You can override using the ``.groups`` argument.



If you look at these plots, you'll see that the bad behavior of Hodges' estimator is temporary for each value of $\psi \neq 0$. As n increases, the estimator eventually behaves like the sample mean. Since the asymptotics don't care about any initial behavior of the sequence, they completely ignore the fact that Hodges' estimator is usually much worse than the sample mean because those problems are "temporary" as far as the behavior at any fixed distribution is concerned.

The issue with that is that is that the problem doesn't actually disappear, it just goes to another distribution! As you can see in our first plot, the region of bad behavior simply moves as we increase n . If you're standing still at a given $\psi \neq 0$ you'd think everything is fine, but if you look across all values of ψ you notice that we're just moving the problem around. That is precisely what the asymptotic arguments ignore: they are incapable of looking at the behavior of each estimator *at multiple distributions simultaneously*. They only consider a single distribution at a time. We call this *pointwise* asymptotic behavior because we're looking at each "point" (distribution) $\mathbb{P}$ in the model $\mathcal{P}$ one-at-a-time. Once we look across all values of ψ at the same time, it's clear that Hodges' estimator is much worse than the sample mean.

Asymptotic linearity is only concerned with pointwise convergence, but to prevent the bad behavior of Hodges' estimator we need some kind of *uniform* convergence: we'd like the sampling distributions of the estimates at each $\mathbb{P}$ to "arrive" or "approach" their limits at the same time. Physically you can imagine flattening a marshmallow: if you push down at a point on the marshmallow (e.g. by squeezing it between your thumb and index finger), the mass will can just go elsewhere. But if you sandwich it between your palms, you ensure that it gets squeezed down everywhere at the same time. That's "uniform" convergence.

Warning

See Section A.1.1 for the definition of uniform convergence of functions. This is a common type of convergence that we often want to generalize for other kinds of objects. In what follows we will present a notion of “uniform” convergence in distribution for random variables.

There’s actually another problem with this estimator that we didn’t capture with our original list of desiderata. We often want to know in advance how big a sample size we might need to get a particular level of confidence in our result (i.e. do a power calculation). If we understand the asymptotic distribution of our estimator under data were generated from a given distribution $\mathbb{P}$, we have everything we need to do this. The problem is that we don’t know what the real distribution of the data are. So what is done in practice is we generate a rough guess $\hat{\mathbb{P}}$ based on domain knowledge or previous data and use that for the power calculation. If $\hat{\mathbb{P}}$ is close to $\mathbb{P}$ we’d expect the required sample sizes to be the same. For this to work, however, we need the limiting asymptotic distribution of our estimates to be “continuous” over distributions: two data distributions that are close should give sampling distributions for a estimator that are close.

For Hodges’ estimator, this is not the case. The limiting distribution is the standard normal no matter how close you get to $\psi = 0$, but suddenly it is the degenerate distribution at that point. The discontinuity in the limiting distribution poses problems for a power calculation: if in reality $\psi = 10^{-9}$ but we did our power calculation assuming $\psi = 0$ (a tiny, tiny error!), we would end up massively underpowering our study.

Estimators like Hodges’ estimator are called *superefficient* for reasons that will be explained in a few sections’ time. In general, superefficient estimators perform better than other estimators on some very small set of distributions at the cost of arbitrarily bad performance that moves around over the other distributions as n increases. We want to rule out that sort of behavior so that we can compare estimators asymptotically without getting burned by their finite-sample behavior.

2.2 Regularity

We’ve identified a problem with just looking at AL estimators: there are always AL estimators that look good in terms of their pointwise asymptotics but are actually bad because they just move the problem around in a way that makes it invisible to the asymptotics.

To rule this out, we’ll impose an addition condition called *regularity*. In a rough sense, regularity usually does two things: it means that limit distributions vary smoothly (so, for example, power calculations are reliable) and that the estimator approaches its limit distribution in a (locally) uniform way across the model (so that asymptotics can’t hide bad performance that moves around the model).

We will start with a toy definition of regularity that we will build up to the formal definition in the next few sections. That will require us to introduce and explain a few abstractions but these will also come in handy later so it's worth doing. For now, though, here's a first draft:

Regularity (version 1)

Definition 2.1.

A Convergence of Random Variables

Throughout chapters 3 and 4 we talk a lot about the convergence of random variables so it helps to understand what we mean by these things. In the definition of asymptotic linearity the term $o_{\mathbb{P}}(n^{-1/2})$ already invokes a specific type of convergence (more on that below).

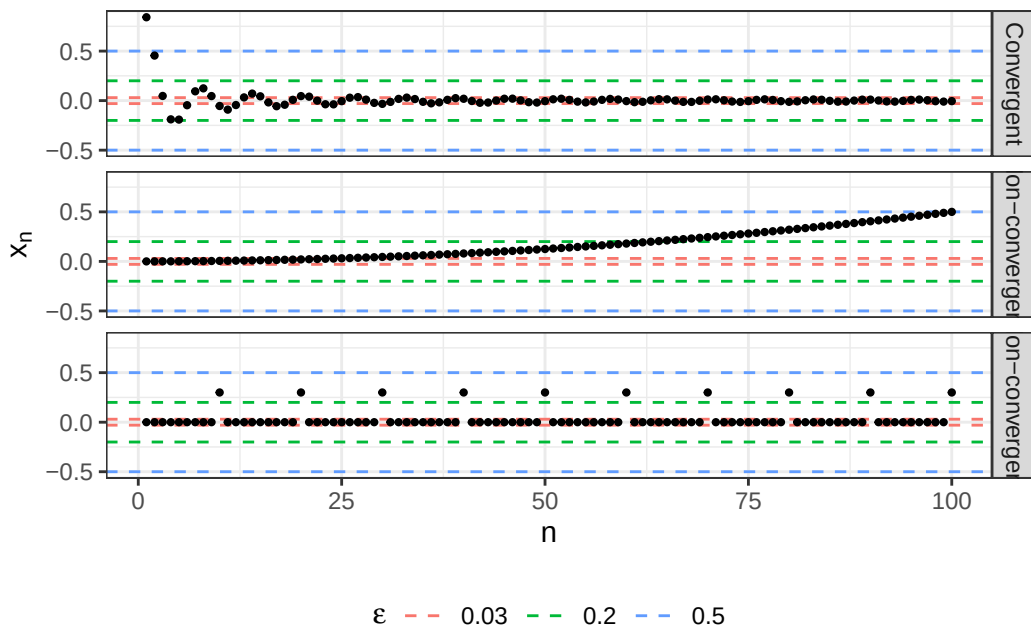
A.1 Sequences and Functions

We'll start with the convergence of a sequence of real numbers. A sequence of real numbers is what it sounds like: an infinite ordered list of numbers. We write a sequence like $[x_1, x_2, x_3, \dots, x_n, \dots]$. Technically we should use something like $\{x_n\}$ to refer to the entire sequence because x_n refers to the n th element, but usually we'll be lazy and just say x_n where it's otherwise clear.

Convergence of Sequences

Definition A.1. A sequence *converges* to the *limit* x if for every $\epsilon > 0$, there is an integer N such that for all $n > N$, we have $|x_n - x| < \epsilon$.

Intuitively, the values in the sequence “approach” the value x , but this is a little misleading because successive elements of the sequence don't necessarily have to be closer to the limit than preceding values. the sequence can oscillate away and towards the limit as long as eventually it becomes trapped forever in the interval $[x - \epsilon, x + \epsilon]$ around the limit. I think this is a better way to think about it: if you can eventually “trap” the sequence in the interval $[x - \epsilon, x + \epsilon]$ no matter how you choose ϵ , then the sequence converges. Convergence is denoted like $x_n \rightarrow x$ or $\lim_{n \rightarrow \infty} x_n = x$.



It's important to understand that the convergence of a sequence does not depend on any of the initial values of a sequence. If I chop off the first 100 values of a sequence which converges, that sequence will still converge. If the original didn't converge, the truncated sequence will still not converge. If I chop off the first billion entries, the same holds. It doesn't matter how many of the initial values I cut because there are infinitely more after them! The point is that convergence is purely a property of how the "tail" of the sequence behaves. And moreover, it doesn't matter how far away from the beginning of the sequence that behavior "kicks in". For that reason, plots like the ones above are actually totally noninformative about the convergence of the sequence because they can't tell us what's happening in the tails. Nonetheless they are a useful teaching tool to build intuition for what convergence looks like.

Functions can be said to converge too. A sequence of functions is just what it sounds like: a neverending list of functions like $\{x^2, 2x^2, 3x^2 \dots\}$.

Pointwise Convergence of Functions

Definition A.2. The sequence of functions f_n *converges pointwise* to a limit f if $f_n(x) \rightarrow f(x)$ as a sequence for every input x .

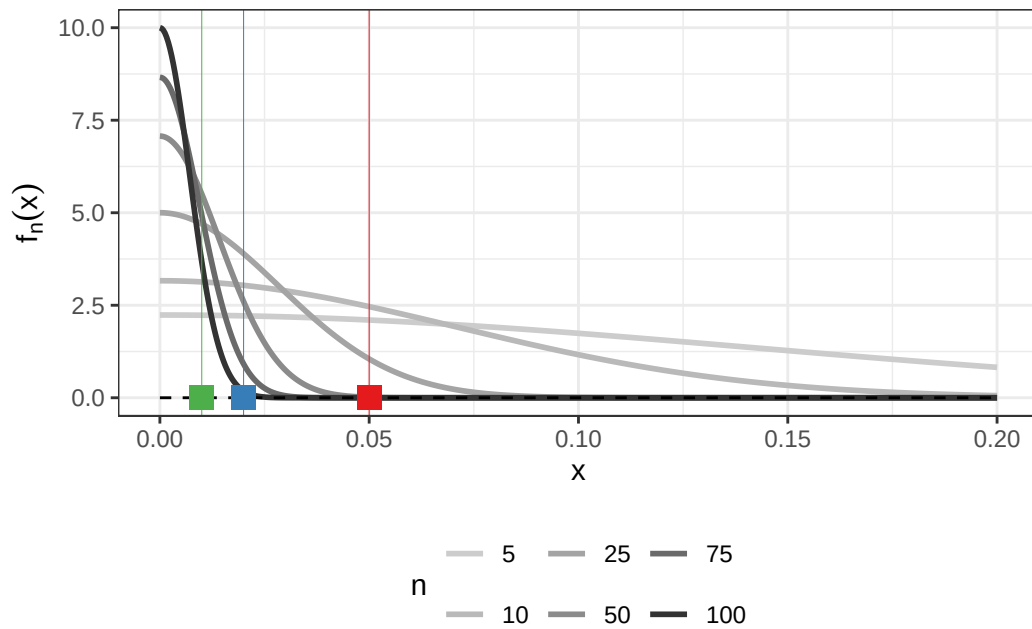
Pointwise convergence is one way to say that the functions f_n are getting closer to f as n increases, but there are actually other ways too! For example, we say that f_n converges to f in mean-squared error if $\int (f_n(x) - f(x))^2 dx \rightarrow 0$ as a sequence. For reasons we'll see later this is sometimes denoted as $f_n \xrightarrow{\mathcal{L}_2} f$.

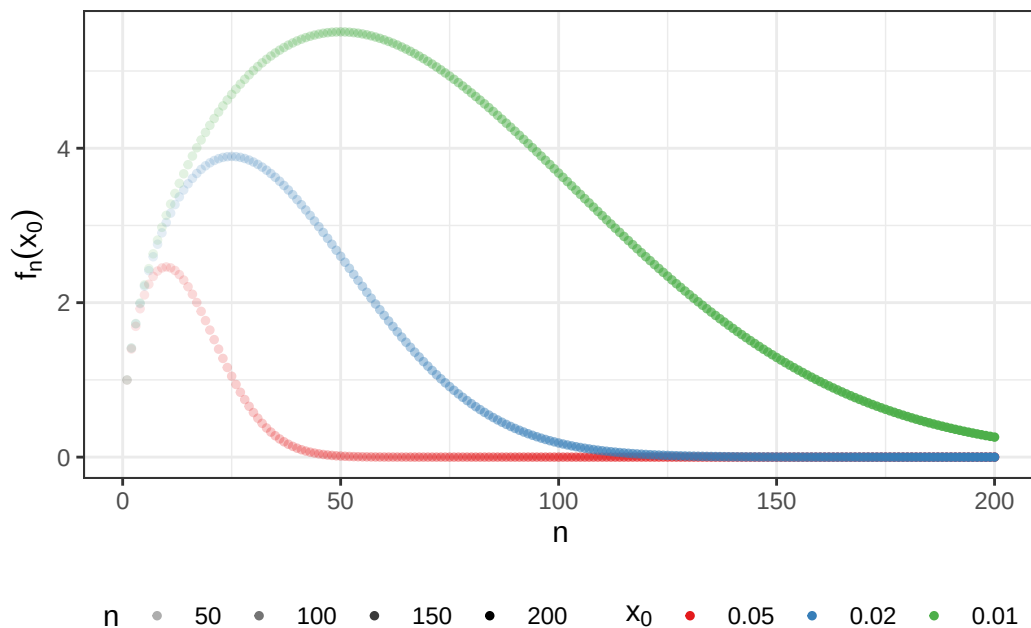
The reason I bring this up is to point out the fact that there are many different senses of “convergence” when you start talking about objects that are more complicated than real numbers. A function can converge to a limit in a mean-squared sense but not in a pointwise sense, and vice versa! Sometimes one type of convergence implies another and sometimes not (without additional assumptions).

Example A.1. In the plots shown here we investigate the convergence of

$$f_n(x) = \sqrt{n}e^{-(nx)^2}$$

on the domain $x \in (0, \infty)$.





As you can see visually, at any point $x_0 \in (0, \infty)$ the sequence of points $a_n = f_n(x_0)$ converges to zero. Therefore we say $f_n \rightarrow 0$ on the domain $(0, \infty)$. However, if you calculate the mean-squared error $\int_0^\infty (f_n - 0)^2 dx$ for this sequence of functions (your favorite LLM can do this for you), you'll see that it stays at a constant, nonzero value of $\sqrt{\pi/8}$. Therefore, while this sequence converges pointwise, it does not converge in a mean-squared sense.

Convergence of sequences and functions is a central topic in real analysis. You can consult an analysis book or lecture series to learn more.

A.1.1 Uniform Convergence

Uniform Convergence of Functions

Definition A.3. f_n converges to f *uniformly* if the largest difference between f and f_n across all values of x at each n converges to 0:

$$\sup_x |f_n(x) - f(x)| \rightarrow 0.$$

Notice that as n changes, the point x at which the difference $|f_n(x) - f(x)|$ is maximized might change!

Example A.2. let $f = 0$ and consider a sequence $f_n(x) = 1(n < x \leq n + 1)$. You can think of the sequence f_n as a bump that moves further and further down the x-axis as n increases. If you stand still at any value x , the bump comes and passes over you and then departs. Therefore, for n large enough (specifically $n > x + 1$), $f_n(x) = 0$ exactly forevermore and thus $f_n \rightarrow f$ pointwise.

That might seem a little weird to you considering that obviously there is still always a bump in f_n that doesn't "go to zero" no matter how big n is. That's part of the intuition that uniform convergence tries to capture. If you look at the largest value of $f_n - f = f_n$, you will find that it is always 1, attained at any value $x \in [n, n + 1]$. Therefore $\sup_x |f_n(x) - f(x)| = 1 \not\rightarrow 0$, meaning this sequence of functions does not converge *uniformly*.

Exercise A.1. In Example A.1 we considered the pointwise convergence of $f_n(x) = \sqrt{n}e^{-(nx)^2}$. Just by looking at the plots, explain why this sequence does not converge uniformly.

Similar concepts pop up in different places across mathematics. in general, when we're looking at some property of the worst-case behavior of a mapping across a variety of inputs simultaneously we usually use the word "uniform" to describe it.

A.2 Random Variables

Finally we get to random variables. When rigorously defined, random variables are actually functions and can converge in many different ways, but most of them will not be important for the rest of this book. For our purposes, we will be able to think about random variables pretty much exclusively as distributions. The type of convergence we will mostly be concerned with is *convergence in distribution*. You can think of a sequence of random variables Z_n as a sequence of distributions. Since all distributions have cumulative distribution functions (CDFs) you can also think of a sequence of CDFs $F_n(z) = F_{Z_n}(z)$. You may lastly also imagine a sequence of densities or mass functions if you like. Not all distributions actually have densities in the way you may be used to thinking about them, but that shouldn't stop you from using densities to build your intuition. I usually picture something like this Joy Division album cover when I think of a sequence of random variables:

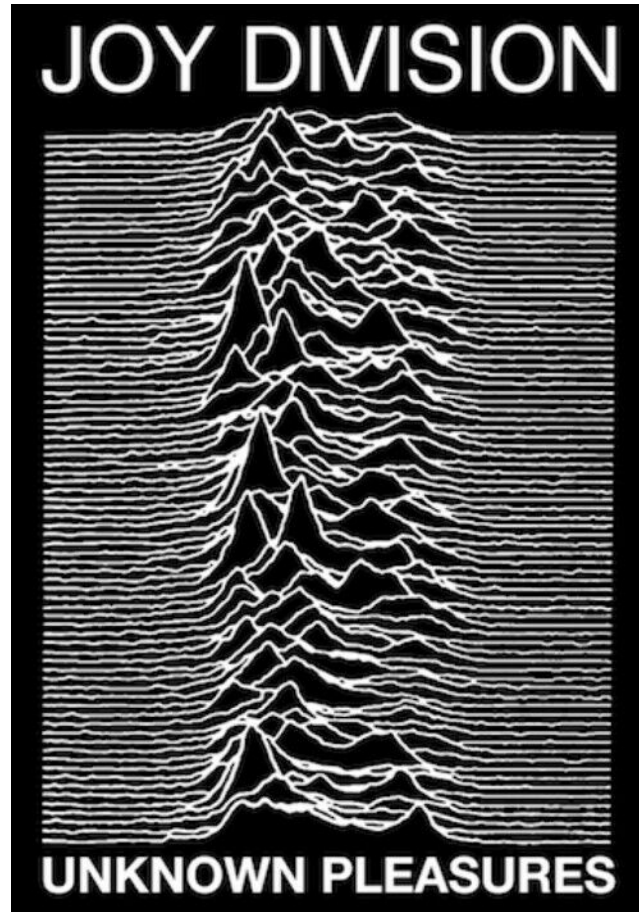


Figure A.1: Joy Division's famous Unknown Pleasures album cover

Convergence in Distribution

Definition A.4. The sequence Z_n converges in distribution (or “converges in law” or “weakly converges”) to a limit Z if and only if the CDFs F_n converge pointwise (as functions) to the CDF F of Z (at all points where F is continuous). We write this as $Z_n \rightsquigarrow Z$

The reason convergence in distribution is so often used is because this is the kind of convergence implied by the central limit theorem (CLT). The (simplest version) of the CLT says that if Z_i are IID random variables with finite variance σ^2 , then

$$\sqrt{n} \left(\frac{1}{n} \sum Z_i - \mathbb{E}[Z] \right) \rightsquigarrow \mathcal{N}(0, \sigma^2).$$

This might look a little complicated but if you rename $X_n = \sqrt{n} \left(\frac{1}{n} \sum Z_i - \mathbb{E}[Z] \right)$ you see that

this is just a particular sequence of random variables (because the Z_i are random variables) and we can read the CLT as saying $X_n \rightsquigarrow \mathcal{N}(0, \sigma^2)$.

As with convergence of real number sequences, you should keep in mind what this definition does *not* imply. It does not imply that the sequence converges “nicely” in the sense that all subsequent elements are always “closer” to the limit than previous elements. It does not tell us anything about *when* the sequence gets “close enough” to its limit: we could get “close” by $n = 10$ or it might take up until $n = 10^{100}$! Convergence says nothing about the behavior at the beginning of the sequence, and the “beginning” could be very, very long.

The other type of convergence of random variables we’ll use is *convergence in probability*. I’ll give you the full definition for completion, but it’s a bit tricky and honestly for the purposes of this book you can get away without really understanding it.

Convergence in Probability

Definition A.5. A sequence of random variables Z_n converges in probability to a random variable Z if $\mathbb{P}\{|Z_n - Z| > \epsilon\} \rightarrow 0$ as a sequence of real numbers for any $\epsilon > 0$. We denote this as $Z \xrightarrow{\mathbb{P}} Z$.

The reason this definition can be tricky is because in general you have to consider the joint distribution of Z_n and Z to calculate the probability invoked in the definition. For this reason, a sequence that converges in distribution might not converge in probability (but the converse is true: convergence in probability is stronger than convergence in distribution).

Thankfully, in this book, the only place we’ll see convergence in probability is when we’re talking about convergence to a fixed constant z (i.e. a degenerate distribution where all the mass is placed at a single value). In this case, the definition simplifies: $Z \xrightarrow{\mathbb{P}} z$ if and only if $Z_n \rightsquigarrow z$. So the only kind of convergence you really need to recognize to read this book is convergence in distribution.

Exercise A.2. Let $Z_n \sim \text{Unif}(0, 1 + 1/n)$. What distribution does this converge to? Draw a picture to sharpen your intuition, then prove the convergence rigorously by taking the pointwise limit of the CDFs.

A.2.1 Little $o_{\mathbb{P}}$

Often in asymptotic statistics we want to say something like

$$Y_n = X_n + U_n \text{ where } U_n \xrightarrow{\mathbb{P}} 0.$$

This is kind of annoying to write! Probabilists and statisticians have come up with a more convenient way to say the same thing:

$$Y_n = X_n + o_{\mathbb{P}}(1).$$

You can read the term $o_{\mathbb{P}}(1)$ as “some sequence of random variables that converges in probability to 0”. The benefit of this is that we don’t even have to give this sequence a name (like U_n , for example) if we don’t care to. Moreover, we’ve hidden away any limit arrows and are left with an equation that we can manipulate algebraically.

This notation is extended to handle cases where we want to say

$$Y_n = X_n + U_n \text{ where } r_n U_n \xrightarrow{\mathbb{P}} 0.$$

The abbreviation for this is $Y_n = X_n + o_{\mathbb{P}}(1/r_n)$. The way I “see” this from the notation is that $r_n U_n = o_{\mathbb{P}}(1)$, so we “divide” by r_n to get

$$U_n = o_{\mathbb{P}}(1)/r_n \equiv o_{\mathbb{P}}(1/r_n).$$

The last equality here defines what we mean when we say some variable U_n is $o_{\mathbb{P}}(1/r_n)$: namely that U_n times r_n converges in probability to 0. When this is the case we say that “ U_n converges at the *rate* $1/r_n$ ”. Sometimes the sequence r_n can also itself be a random variable R_n , but the meaning is the same: if you see $U_n = o_{\mathbb{P}}(1/R_n)$ it means that $U_n R_n \xrightarrow{\mathbb{P}} 0$.

The intuition for “rates” is that often we’re interested in random variables that go to 0 so quickly that even if they get blown up by some increasing sequence, the product still goes to 0. For example, if some variable U_n is $o_{\mathbb{P}}(n^{-1/2})$, that means that $\sqrt{n}U_n$ still goes to 0, even though $\sqrt{n}$ is sequence that gets bigger and bigger. The rate r_n ($r_n = n^{1/2}$ in this example) is the factor by which we can afford to “blow up” the sequence of variables U_n and still have it converge to 0. Another way to think of it is that $U_n = o_{\mathbb{P}}(1/r_n)$ means that $U_n < C/r_n$ with high probability (for some constant C and large enough n). Notice that this is a strict inequality: more on that later.

Example A.3. Let’s say that $Z_n \sim \mathcal{N}(0, \sigma_n^2)$ where $\sigma_n = n^{-1/2}$. This is what we’d expect the large-sample sampling distribution of $\hat{\psi} - \psi$ to look like for any asymptotically linear estimator (see Definition 1.1).

Below, you can see some density plots for this sequence and how they change if they are blown up by different sequences. In the top two rows, the scaling factor is 1 (not increasing) or $n^{1/4}$ (increasing very slowly) and you can see that the densities are still narrowing as n increases, albeit a little more slowly when blown up by $n^{1/4}$. However, in the third row the sequence is blown up by the moderately increasing factor $n^{1/2}$. You can see this is enough to perfectly “stabilize” the distribution: it neither narrows nor widens as n increases. Lastly, in final row

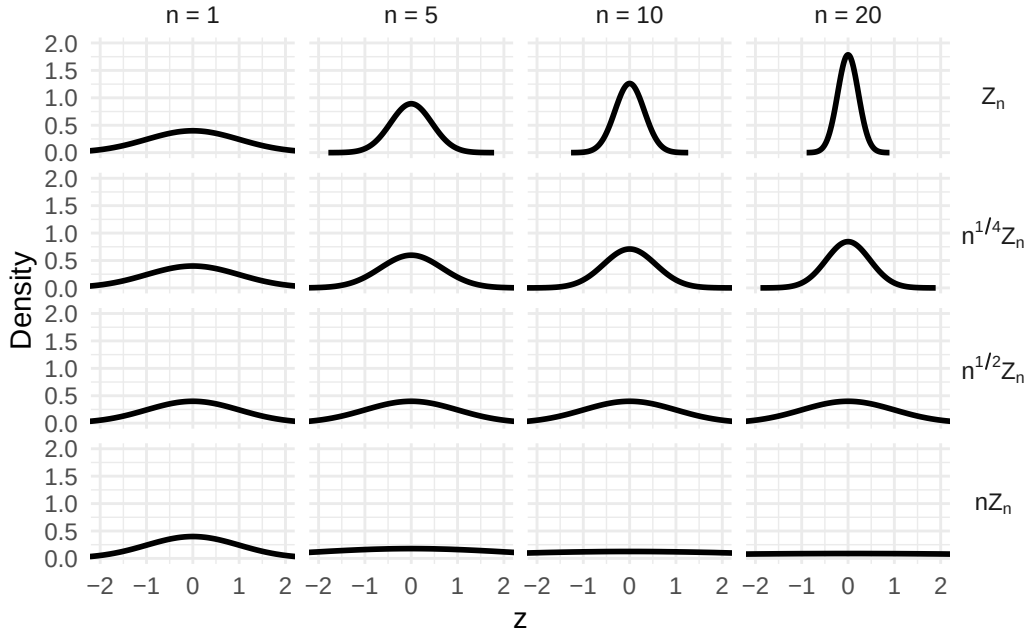


Figure A.2: Scaled normal densities.

we've blown up Z_n by the massive factor n and you can see the distribution widens very quickly as n increases.

Since we can see $Z_n \xrightarrow{\mathbb{P}} 0$ we could write $Z_n = o_{\mathbb{P}}(1)$. Similarly, since we see $n^{1/2}Z_n \xrightarrow{\mathbb{P}} 0$ we could also write $Z_n = o_{\mathbb{P}}(n^{-1/4})$, which is a stronger statement. However, we *cannot* write $Z_n = o_{\mathbb{P}}(n^{-1/2})$ because $n^{1/2}Z_n$ is not collapsing to zero. Similarly we cannot write $Z_n = o_{\mathbb{P}}(n^{-1})$.

Exercise A.3. Let $U_n \sim \text{Unif}(0, 1/n)$. Is $U_n = o_{\mathbb{P}}(1)$? Is $U_n = o_{\mathbb{P}}(1/n)$?

Writing $U_n = o_{\mathbb{P}}(1/r_n)$ tells us something about the sequence U_n without entirely specifying what that sequence is. The use of the equals sign is a big abuse of notation because obviously two sequences can be “equal” to the same $o_{\mathbb{P}}(1/r_n)$ without being equal to each other. Moreover anything that is $o_{\mathbb{P}}(n^{-1/2})$ is also $o_{\mathbb{P}}(1)$ (something that converges to zero when blown up still converges to zero if it is not), but not the other way around. The utility of the $o_{\mathbb{P}}$ notation is that it lets us show some important property of a sequence of random variables within the context of an equation. It just saves a line or two of exposition. In that sense it's not that different from other conventions like consistently using capital letters for random variables: even without extra exposition you already know an important property of the objects you're looking at.

The most common use of this notation is to tell the reader that some term is “negligible” (in large samples) relative to some other term of greater analytic importance. For example, let’s say that $X_n \sim \mathcal{N}(0, \sigma^2/n)$ and I know that $Y_n = X_n + U_n$. By the definition of X_n , we know that

$$\sqrt{n}X_n \sim \mathcal{N}(0, \sigma^2).$$

This does *not* converge to zero: it stays stable at this normal distribution with “spread” σ even as n increases. If we can show that $\sqrt{n}U_n \xrightarrow{\mathbb{P}} 0$ (also written $U_n = o_{\mathbb{P}}(n^{-1/2})$), then we would know that U_n is “negligible” compared to X_n in large samples: U_n converges to 0 even when blown up by $\sqrt{n}$ whereas X_n doesn’t. Therefore, asymptotically, $Y_n \approx X_n$. To make this a little more precise we would write $Y_n = X_n + o_{\mathbb{P}}(n^{-1/2})$ to indicate “how quickly” this approximation kicks in.

Alternatively, if you saw $Y_n = X_n + o_{\mathbb{P}}(n^{-1/2})$ without knowledge of the distribution of X_n then you would know that $Y_n \approx X_n$ asymptotically only if X_n converges more slowly (if at all) to zero than $n^{-1/2}$. Alternatively, you would know that Y_n converges to zero at an $n^{-1/2}$ rate if and only if $X_n = o_{\mathbb{P}}(n^{-1/2})$.

Usually the most important part of a rate will be some polynomial factor like $n^{1/2}$. You should remember that the larger the exponent, the faster a polynomial grows. If you have a negative exponent, the larger in magnitude it is the faster it decays. A term that is $o_{\mathbb{P}}(n^{-1})$ decays very quickly, while $o_{\mathbb{P}}(n^{-1/2})$ decays more slowly and $o_{\mathbb{P}}(n^{-1/3})$ more slowly still.

A.2.2 Big $O_{\mathbb{P}}$

The last thing to know about is the “big oh-p” notation. If $U_n = O_{\mathbb{P}}(1/r_n)$, then it means that $r_n U_n$ is *stochastically bounded* (also called *asymptotically tight*). Essentially, a sequence that is stochastically bounded does not diverge. For a sequence of random variables to “not diverge” we mean that its probability mass mostly concentrates in some region $[-b, b]$ (but doesn’t necessarily accumulate at a single point). We’ll formalize this in a minute.

For example, let’s say that the random variables Z_n converge to a normal distribution: since the limit is stable, we’d like to say that this sequence does not diverge. But because the normal is unbounded, there is no region $[-b, b]$ that contains all of the mass of the limit distribution. However, for a given tolerance ϵ , we can always find a region that contains $1 - \epsilon$ percent of the mass.

To rigorously define stochastic bounding we’ll just generalize this insight.

Stochastic Bounding

Definition A.6. A sequence of random variables Z_n is stochastically bounded if for any given tolerance ϵ we can find a region $[-b, b]$ so that $\lim_n \mathbb{P}(|Z_n| < b) > 1 - \epsilon$.

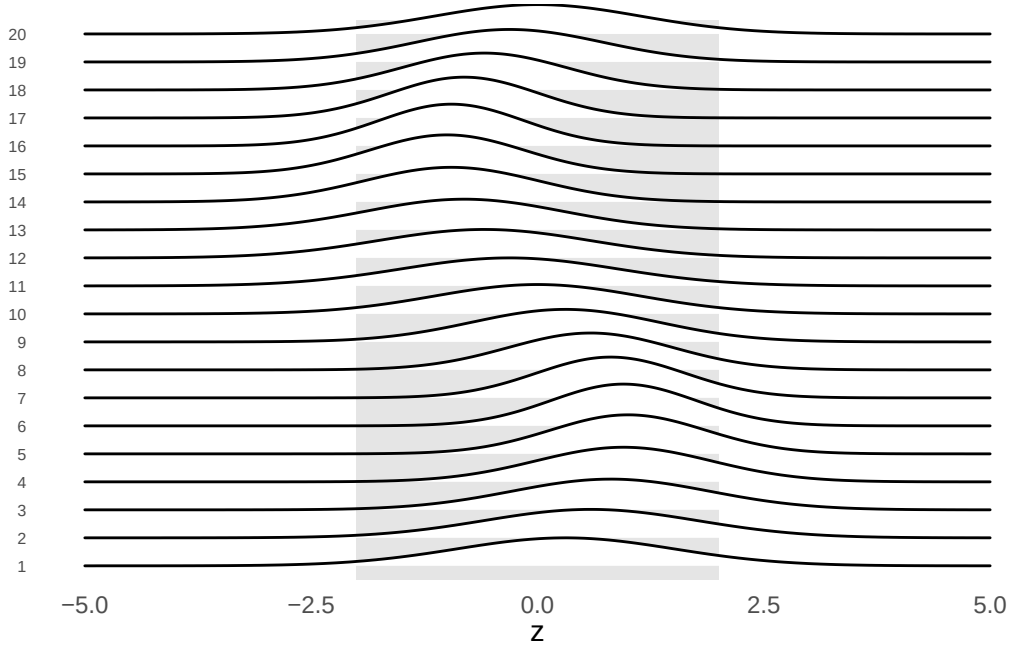


Figure A.3: The sequence of random variables $\mathcal{N}(\sin(n), \sin(n) + 2)$ does not converge in distribution yet is still stochastically bounded.

Saying a sequence is stochastically bounded is only *slightly* weaker than saying it converges to zero. If we multiply a stochastically bounded sequence by any sequence that converges to zero, however slowly, the resulting sequence will converge to zero. On the other hand, if we multiply it by any sequence that diverges, however slowly, then the resulting sequence also diverges. A stochastically bounded sequence that does not converge to zero is therefore on the knife's edge between convergence (to zero) and divergence. Another way to think of it is that $U_n = O_{\mathbb{P}}(1/r_n)$ means that $U_n \leq C/r_n$ with high probability (for a constant C and large enough n). Notice that for big O it's less than *or equal to* whereas for little o it's a strict inequality.

Any sequence of random variables that converges in distribution is stochastically bounded. This makes sense if we're thinking of stochastic boundedness as “almost” convergence to zero: if the sequence converges to a (nonzero) stable limit distribution, any additional shrinking will collapse it to zero.

Example A.4. The CLT says that under a variety of conditions

$$\sqrt{n} \underbrace{\left(\frac{1}{n} \sum Z_i - \mathbb{E}[Z] \right)}_{X_n} \rightsquigarrow \mathcal{N}(0, \sigma^2)$$

Since the demeaned sample average X_n converges to a stable distribution when scaled by $\sqrt{n}$, it does not diverge. Therefore we can say $X_n = O_{\mathbb{P}}(n^{-1/2})$. However, notice that this limit is not zero. Therefore $X_n \neq o_{\mathbb{P}}(n^{-1/2})$.

Often the big O is used to indicate that r_n is *exactly* the rate that “stabilizes” the asymptotic distribution of U_n so that is neither converging to zero nor diverging. Any rate that is slightly slower (like $n^{1/2-\epsilon}$) and $r_n U_n$ would go to zero. Anything slightly faster (like $n^{1/2+\epsilon}$) and it would blow up.

Example A.5. Recall the example above where $Z_n = \mathcal{N}(0, \sigma_n^2)$ where $\sigma_n = n^{-1/2}$. In this case we saw that $n^{1/2}Z_n$ does *not* converge to zero in probability. However, $n^{1/2}Z_n$ *is* stochastically bounded because it converges to a stable limit distribution (in this case $N(0, 1)$). Therefore while we cannot write $Z_n = o_{\mathbb{P}}(n^{-1/2})$, we *can* write $Z_n = O_{\mathbb{P}}(n^{-1/2})$ with the big O .

Technically something that is $O_{\mathbb{P}}(1/r_n)$ might also be $o_{\mathbb{P}}(1/r_n)$ (big O means less than or equal to, which *could* be strictly less). But usually for given rates $1/a_n < 1/b_n$ (e.g. $n^{-1/2} < n^{-1/3}$) you should roughly understand that $o_{\mathbb{P}}(1/a_n) < O_{\mathbb{P}}(1/a_n) < o_{\mathbb{P}}(1/b_n)$.

You can learn more about this notation in a number of online resources (google “little oP” or “little oh p”) or in chapter 2 of Asymptotic Statistics (van der Vaart). A thorough understanding requires a grasp of convergence in probability.

B $\mathcal{L}_2$ Space

It will help you a lot to learn efficiency theory if you have a solid understanding of $\mathcal{L}_2$ and a few basic ideas from functional analysis. Many students of statistics are missing this background so I'll try and give you the necessary intuition here.

B.1 Vector Spaces

The first thing to understand is that functions are a kind of vector. This might make no sense to you if you're only familiar with the "vectors" you learned about in your first linear algebra class, which are finite-dimensional objects like $x = [x_1, x_2, x_3]$. But the formal definition of "vector" is actually much broader than this.

Real Vector Space

Definition B.1. A (real) *vector space* is a set $\mathcal{V}$ that satisfies the following properties:

- 1) **Closure under scalar multiplication:** Multiplication between elements of $\mathcal{V}$ and real numbers is defined. For all $v \in \mathcal{V}$ and real numbers c , the product cv is also in $\mathcal{V}$.
- 2) **Closure under addition:** Addition between elements of $\mathcal{V}$ are defined. For any two elements $u, v \in \mathcal{V}$, their sum $u + v$ is again in $\mathcal{V}$.

The elements $v \in \mathcal{V}$ are what we call vectors.

Technically this is a special case of vector space called a *real* vector space (one built on the real scalar field) but for our purposes that will suffice.

Exercise B.1. Why is the interval $[0, 1]$ not a vector space? Why is the set of all rational numbers not a (real) vector space?

Our usual (finite-dimensional) vectors are a special case of this. The vectors in the space $\mathbb{R}^d$ are all the ordered tuples of d real numbers. If $x = [x_1, \dots, x_d]$ and $y = [y_1, \dots, y_d]$, we define scalar multiplication as $cx = [cx_1 \dots cx_d]$ and $x + y = [x_1 + y_1, \dots, x_d + y_d]$. Clearly if I add two of these objects together or scale one of them using these definitions I get another vector in the same set, which means that we can call this set a vector space and its elements vectors.

Technically we should also include the definitions of scaling and addition as part of the vector space since the set by itself doesn't naturally come "equipped" with these definitions.

If you were to "code" a vector space, you might do it something like this:

```
class Vector:

    def __init__(self::Vector):
        ... # define what kind of object the "vector" is

    def __add__(self::Vector, other::Vector):
        ... # define how to add two vectors so you can evaluate v + u

    def __mult__(self::Vector, c::Number):
        ... # define how to scale a vector by a number so you can evaluate c*v

VectorSpace = Set( # a set containing various Vectors (usually infinite)
    Vector(...),
    Vector(...),
    ...
)
```

This has no practical value, I'm just trying to help you wrap your head around the ideas in case you are more familiar with programming than you are with abstract mathematics.

The definitions I gave for addition and scaling may seem so natural to you that you don't even think about them but in fact they are just one specific way we could define these operations for finite dimensional ordered tuples. You can check that the given definitions satisfy the usual properties of "addition" and "multiplication" (e.g. $a + (b + c) = (a + b) + c$) so it is sensible to call them by those names.

You can make a vector space out of pretty much anything upon which you can define some kind of scaling and addition. For example, consider the space of all functions that map real numbers to real numbers. If you give me a function f , let me define multiplication of a function by a scalar c to produce another function in the following way: cf is a new function that maps reals to reals where the output $(cf)(x)$ is given by $c \cdot f(x)$. Similarly, define addition of two functions as $(f + g)(x) = f(x) + g(x)$, i.e. the function sum $f + g$ is the unique function which when evaluated at any x gives the value $f(x) + g(x)$. Again this might seem so natural as to not need explanation (or to be made confusing by explanation) but you should be aware that these are man-made definitions of "addition" and "scaling" for functions rather than numbers. Equipped with these operations, the set of all functions from reals to reals becomes a vector space and the functions within it are the vectors.

B.1.1 Functions as Vectors

In linear algebra or vector calculus you may have built some intuition about finite-dimensional vectors by imagining them as “arrows” from the origin to a particular point in 3D (or higher dimensional) space. This helps us reason about vectors geometrically using our physical sense of space. For example, you have probably seen the usual definition of the “length” of a vector (the *norm*) for d -dimensional vectors: $\|x\| = \sqrt{\sum^d x_i^2}$. You probably also know the *inner product* (dot product) $x \cdot y = \sum^d x_i y_i$ (more generally written $\langle x, y \rangle$) and may recall that the inner product measures the *angle* between two vectors according to the law of cosines: $\cos(\theta_{x,y}) = \frac{x \cdot y}{\|x\| \|y\|}$. If the inner product is zero, the two vectors are *orthogonal*. You may also recall that the norm is actually defined from the inner product, i.e. $\|x\|^2 = x \cdot x$.

Seeing functions as vectors (“arrows”) helps us apply similar geometrical reasoning. We can talk about the “length” of a function or the “angle” between two functions. We just need to come up with definitions for those things that satisfy our expectations for how geometry should work. But before doing that I want to give you some more intuition to help you “see” functions as vectors outside of an abstract definition of a vector space.

I like to think of a function as a (continuously) infinite list of numbers like this:

$$f = [\dots f(-1), \dots f(0), \dots f(1.1), \dots f(\pi), \dots].$$

Each “element” of this function/vector is the *value* that the function takes at a particular argument x . Instead of using function evaluation notation with the parentheses I could instead write it with the arguments as *indices*:

$$f = [\dots f_{-1}, \dots f_0, \dots f_{1.1}, \dots f_{\pi}, \dots].$$

This looks the same as a d -dimensional vector $x = [x_1, \dots x_d]$ except that now the set of indices is e.g. all real numbers $\mathbb{R}$ instead of the finite set $\{1, \dots d\}$. To me, this perspective makes functions “feel” like very long, densely packed vectors. Since the coordinate axes of a standard d -dimensional vector space correspond to the vector indices $\{1, 2, \dots d\}$, the “axes” of a vector space of functions might be thought of as corresponding to all the possible arguments x of the function(s). Different functions evaluate to different values along each of these (continuously infinite) different axes. Thinking this way shows that definitions of scaling and addition now work exactly the same “elementwise” way for finite-dimensional vectors and functions.

To complete the picture it’s also helpful to notice that finite-dimensional vectors can also be written as functions. Instead of writing e.g. $x = [x_1, x_2, x_3]$ I could say that x is a function on the domain $\{1, 2, 3\}$ such that $x(1) = x_1$, $x(2) = x_2$, and $x(3) = x_3$. Indices and arguments are essentially two different ways to view or construct the same thing. The difference between a function (e.g. on the reals) and a finite-dimensional vector is just their domains if you want to see them both as functions (or their index sets if you want to see them both as “vectors”).

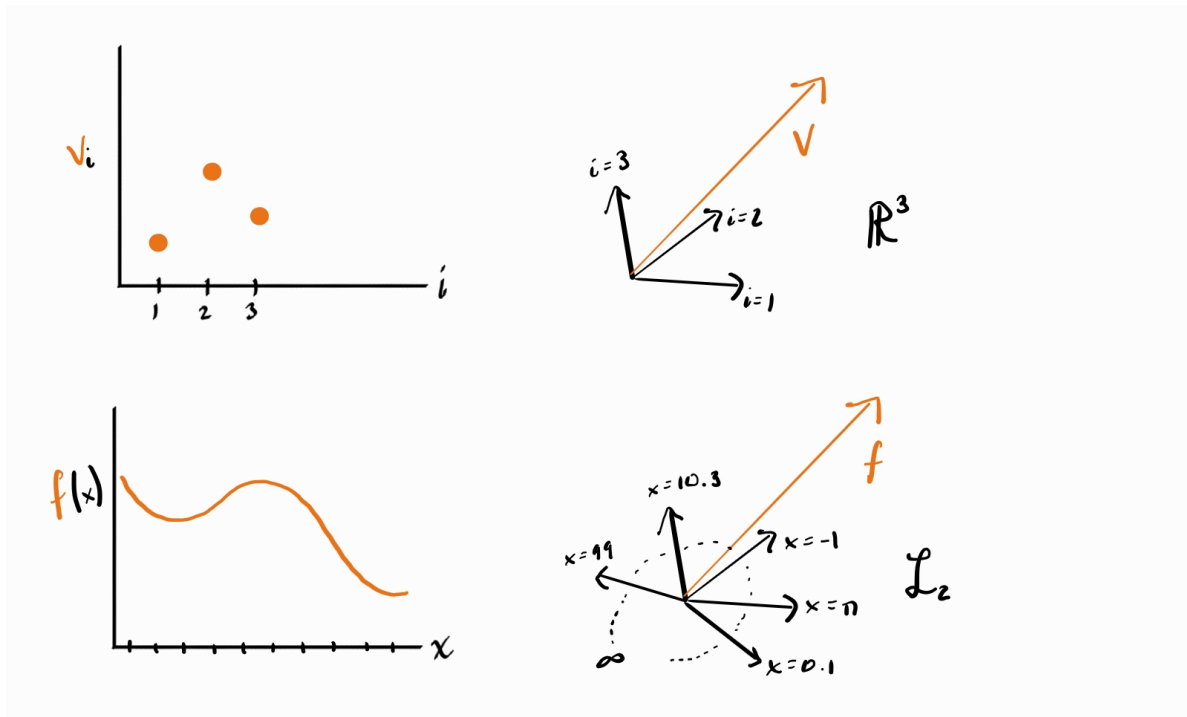


Figure B.1: Functions as vectors and vice-versa

Hopefully at this point the visual intuition doesn't seem so crazy to you, even when you're working with infinite dimensional spaces. But if you're having trouble, follow the advice of computer scientist Geoffrey Hinton: "to deal with a 14-dimensional space, visualize a 3-D space and say 'fourteen' to yourself very loudly. Everyone does it."

It's funny, but this kind of thinking is a staple of mathematics. We all use stupid pictures or imaginations grounded in our physical experience which we know are "wrong" in some way (e.g. 3 dimensions instead of 14). It's the only way to build intuition! The important thing is knowing that the pictures are wrong in some ways. But the more wrong pictures you have (e.g. from different "perspectives"), the better, since you can switch to a picture that "correctly" illustrates the phenomenon of interest.

B.2 $\mathcal{L}_2$ Space

Now that we've built that intuition we can start to formalize some of our geometric ideas. First up: what is the "length" of a function? If the usual length of a finite vector is determined by the (square root of the) sum of the squares of its elements, then we could extend that definition to a function. The only problem is that $\sum_{x \in \mathbb{R}} f(x)^2$ is not well-defined: the finite sum is not

appropriate to deal with an index set that is continuously infinite. But we do already have a “continuously infinite sum”: the integral. Thus we might define

$$\|f\|^2 = \int_{-\infty}^{\infty} f(x)^2 \, dx$$

to be our norm. We call this the $\mathcal{L}_2$ norm (there are other choices; the “2” is for the square inside the integral). Although there are other possible norms, the $\mathcal{L}_2$ norm is nice because it is compatible with an inner product, namely

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)g(x) \, dx.$$

You should again think of this as the continuously-summed version of the finite-dimensional dot product. Hopefully the intuition from thinking of functions as long, dense “vectors” is helpful to you here. If you replace the integral with a sum and the function call notation with an index you will see how this norm and inner product are analogous to the finite-dimensional versions you’re more familiar with.

There are some functions which have infinite or undefined $\mathcal{L}_2$ norm. For example, $f(x) = x^2$ would have infinite norm since (for now) we’re integrating over the entire number line and the area under the curve of $(x^2)^2$ goes off to infinity. To preserve the geometric intuition of our vector space, we cannot include these functions (the same way there is no finite-dimensional vector that has infinite norm). The vector space $\mathcal{L}_2$ (read: “el-two”) is therefore defined as the set of all functions with finite $\mathcal{L}_2$ norm (not quite but see below). Sometimes this is also called the space of “square-integrable” functions. You can check that this is indeed a vector space, i.e. that two functions with finite square integrals sum to a function with a finite square integral and that a function with a finite square integral scales to another function with finite square integral.

Example B.1. Consider the functions $f = 1_{[0,1]}$ and $g = 1_{[2,3]}$. According to our inner product, these two functions are orthogonal (compute the integral and see). If this is not obvious to you at first, consider the finite-dimensional vectors $[0, 1]$ and $[1, 0]$: hopefully these are “obviously” orthogonal to you: you can visualize them as unit vectors along the two axes of $\mathbb{R}^2$. Our two functions are no different: f is nonzero along the “axes” corresponding to the indices in $[0, 1]$ while g is nonzero only along “axes” $[2, 3]$. If you imagine them as long dense “vectors” you’ll see that f and g have ones in different places, the same way that $[0, 1]$ and $[1, 0]$ do.

A further wrinkle is that until now I’ve used the standard Riemman integral (with the differential dx) to define $\mathcal{L}_2$ but that is not quite right. In reality it’s slightly more complicated. We formally define $\mathcal{L}_2(\mathbb{P})$ to be the space of $\mathbb{P}$ -measureable, square-Lebesgue-integrable functions. The measurability is a technical condition that rules out some extremely esoteric corner cases that you don’t need to worry about: read up on measure theory if you want to learn more.

Although the $\mathcal{L}$ in $\mathcal{L}_2$ stands for “Lebesgue”, the “Lebesgue” part is also not terribly important. The Lebesgue integral is a formalization of the integral that can integrate strictly more functions than the Riemann integral, but any function that is Riemann integrable Lebesgue-integrates to the same value so it doesn’t make much difference in terms of intuition.

The important part is the $\mathbb{P}$: one of the strengths of the Lebesgue integral is that it can be defined with respect to any “measure” (probability distributions are formally mathematical objects called measures). When you integrate “with respect to a distribution $\mathbb{P}$ ” you write $\int f \, d\mathbb{P}$ or $\int f(z) \, d\mathbb{P}(z)$ to indicate more explicitly that $\mathbb{P}$ is a distribution over Z and that f takes arguments in the range of Z . If you’re not familiar with measure theory (I don’t expect you to be), when you see an integral with respect to a continuous distribution $\mathbb{P}(Z)$ with density function $p(z)$ you should read it as

$$\int f(z) \, d\mathbb{P}(z) = \int f(z) \, p(z) \, dz = \mathbb{E}_{\mathbb{P}}[f(Z)]$$

In other words, you can think of the differential $d\mathbb{P}$ as weighting different parts of the domain of z in the integral by the probability of z falling into them. Integrating a function with respect to a distribution over Z is the same as taking its expectation over the distribution of Z .

However, $\mathcal{L}_2(\mathbb{P})$ spaces can also be defined when $\mathbb{P}(Z)$ is a discrete distribution (with probability mass function $p(z)$). In that case, $\int f(z) \, d\mathbb{P}(z) = \sum f(z) \, p(z) = \mathbb{E}_{\mathbb{P}}[f(Z)]$. The integral is also defined for distributions that are neither strictly continuous or discrete (e.g. Z is 0 half the time and a standard normal the other half; think of a normal density with an infinite “spike” of mass $1/2$ at 0). You don’t have to worry about this to get all the intuition you need for this book.

You’ll notice that some functions will be square-integrable under some distributions but not under others. There isn’t really one single $\mathcal{L}_2$ space when we’re working with probability distributions. In the wild, if you see “ $\mathcal{L}_2$ ”, that’s short for the $\mathcal{L}_2$ space based on the Lebesgue measure: roughly speaking, the integrals used to define the norm and inner product don’t use any weights on the real number line. But we’ll always be working with $\mathcal{L}_2(\mathbb{P})$ for some particular $\mathbb{P}$. Nonetheless I may sometimes refer to “ $\mathcal{L}_2$ ” generically. If I do that, it’s just as a shorthand for $\mathcal{L}_2(\mathbb{P})$ for a specific distribution that is clear from context or for any generic distribution. Similarly, if we write a norm or inner product we should technically always specify the relevant space, i.e. $\|f\|_{\mathcal{L}_2(\mathbb{P})}$ or $\langle f, g \rangle_{\mathcal{L}_2(\mathbb{P})}$. However, to minimize notational burden we often leave off the details of the space and/or base measure when they are clear from context.

Example B.2. $f(z) = z^{-1/2}$ is square-integrable if $Z \sim \text{Unif}[2, 3]$ but not if $Z \sim \text{Unif}[0, 1]$.

The very last detail we need to attend to is that two different (pointwise) functions need to be considered “equivalent” in $\mathcal{L}_2$. This is a very minor detail that you can quickly forget about for most purposes but I’ll go over it here for completeness.

In finite-dimensional vector spaces, if you have two vectors v and u and $\|v - u\| = 0$, then $v = u$. We'd like this to work in $\mathcal{L}_2$ but it doesn't without a little extra massaging.

Exercise B.2. Take $f(z) = 1(z = 0)$ and $g(z) = 0$. Argue that under any distribution $\mathbb{P}$ which has a traditional density function we get $\|f - g\| = 0$ for the $\mathcal{L}_2(\mathbb{P})$ norm.

The problem highlighted by Exercise B.2 is that the $\mathcal{L}_2$ distance “ignores” pointwise differences (in fact any countable number of them). Therefore in order to get our vector space to work the same way as $\mathbb{R}^d$ we need to say that f and g are effectively the same function as far as $\mathcal{L}_2(\mathbb{P})$ is concerned. Formally, we say that the elements (vectors) of $\mathcal{L}_2(\mathbb{P})$ are not functions themselves, but mutually exclusive groups of functions that are equivalent to each other for purposes of computing norms or inner products. We call these groups *equivalence classes*. Again, this is not really that critical to understand and doesn't have many implications as far as this book is concerned. Practically speaking, everyone considers *functions* to be the vector elements of $\mathcal{L}_2$ spaces, not equivalence classes of functions.

Now we can finally state our “official” definition of $\mathcal{L}_2(\mathbb{P})$:

$\mathcal{L}_2(\mathbb{P})$

Definition B.2. Given a distribution $\mathbb{P}$, The vector space $\mathcal{L}_2(\mathbb{P})$ is the set of equivalence classes of functions $[f] = \{g : \mathbb{P}(g - f)^2 = 0\}$ for which the norm $\|f\|^2 = \mathbb{P}f^2$ is defined and finite. We equip this set with the inner product $\langle f, g \rangle = \mathbb{P}fg$ which induces the norm.

Exercise B.3. Let $\mathbb{P}(Z)$ be such that $Z \sim \text{Unif}(0, 2\pi)$. Calculate the $\mathcal{L}_2(\mathbb{P})$ inner product between $\cos(z)$ and $\sin(z)$. Based on the law of cosines, what does that mean the “angle” is between these functions?

The (squared) $\mathcal{L}_2(\mathbb{P})$ norm has an important probabilistic interpretation: it represents the (uncentered) second moment of $f(Z)$:

$$\|f\|^2 = \mathbb{E}[f(Z)^2].$$

For functions that have zero mean ($\mathbb{E}[f(Z)] = 0$), this is the variance $\|f\|^2 = \mathbb{V}[f(Z)]$. Therefore the “length” of a (zero mean) random variable is nothing other than its standard deviation.

Exercise B.4. Let a random variable Z and functions f, g be such that $\mathbb{E}[f(Z)] = \mathbb{E}[g(Z)] = 0$. Use the definitions of expectation and covariance to show that $\mathbb{E}[f(Z)^2] = \|f\|^2$ and $\mathbb{C}[f(Z), g(Z)] = \langle f, g \rangle$ where the norm and inner product are with respect to $\mathcal{L}_2(\mathbb{P})$.

$\mathcal{L}_2$ is the canonical example of a *Hilbert space*, which is just an inner product space that contains its limits. So if you see that term anywhere and aren't familiar with it, just think of $\mathcal{L}_2$ or $\mathbb{R}^d$. We won't look at Hilbert spaces besides these in this book but nonetheless I'll use the term "Hilbert space" just so I don't have to write out " $\mathcal{L}_2$ or $\mathbb{R}^d$ " every time.

These topics are part of the foundations of measure theory and functional analysis. If you're curious about function spaces and integrals you should check out a book or take a course on one of these subfields (after some introductory real analysis). My recommendations for self-study are in the introduction to this book.

B.3 Triangles in Inner Product Spaces

In this section we'll investigate some properties of Hilbert spaces that come from simple geometric arguments about triangles. These properties are generic to all Hilbert spaces so when I say "vector" in the following I mean it in the general sense that could indicate a function, not just a finite-dimensional vector. To prove these properties I'll use the general definition of an inner product.

Inner Product

Definition B.3. A (real) inner product is any function $f(u, v)$ of two vectors that satisfies:

1. **Symmetry:** $f(u, v) = f(v, u)$
2. **Linearity:** $f(cv + w, u) = cf(v, u) + f(w, u)$ for vectors u, v, w and scalar c
3. **Positive-Definiteness:** $f(v, v) > 0$ for all $v \neq 0$

We use the special notation $\langle u, v \rangle$ to denote an inner product. If there are multiple inner products in play we use a subscript on the right bracket to distinguish them.

You can check that the usual dot product and the $\mathcal{L}_2(\mathbb{P})$ inner product we defined above satisfy these conditions. In what follows below I'll also use the fact that the norm is induced by the inner product: $\|v\|^2 = \langle v, v \rangle$. As a consequence of this and the definition above we have the simple corollaries $\|cv\| = c\|v\|$ and $\|v\| = 0 \implies v = 0$.

B.3.1 Pythagorean Theorem

You may recall the Pythagorean theorem from geometry. For a right triangle with two right-angle adjacent sides of length a and b , the length of the hypotenuse c is given by

$$c^2 = a^2 + b^2$$

We say two vectors are *orthogonal* (which we write $u \perp v$) if their inner product is zero: $\langle u, v \rangle = 0$. This is another way to formalize the notion of there being a right angle between vectors. Recall that you can think of $v - u$ as (the translation to the origin of) the “arrow” going from the point u to the point v . Therefore when $u \perp v$, you can think of v and u are the “sides” of a right triangle whose hypotenuse is $v - u$. In that case, the pythagorean theorem tell us that $\|v - u\|^2 = \|v\|^2 + \|u\|^2$. In $\mathcal{L}_2$ and general Hilbert spaces things work the same way.

General Pythagorean Theorem

Theorem B.1. *In any Hilbert space, if f and g are orthogonal, then $\|f - g\|^2 = \|f\|^2 + \|g\|^2$.*

Proof of Theorem B.1

Recall that $\|f\|^2 = \langle f, f \rangle$ and that $f \perp g$ means $\langle f, g \rangle = 0$. Then,

$$\begin{aligned}\|f - g\|^2 &= \langle f - g, f - g \rangle \\ &= \langle f, f \rangle - 2\langle f, g \rangle + \langle g, g \rangle \\ &= \|f\|^2 - 0 + \|g\|^2.\end{aligned}$$

Exercise B.5. Let $\mathbb{P}(Z)$ be such that $Z \sim \text{Unif}(0, 10)$. Let $f(z) = 1(0 < z < 1)$ and let $g(z) = 1(5 < z < 6)$. Check that $f \perp g$ in $\mathcal{L}_2(\mathbb{P})$ and verify that $\|f - g\|^2 = \|f\|^2 + \|g\|^2$.

B.3.2 Projection

The Pythagorean theorem often comes up in the context of projection. Recall that in finite-dimensional vector spaces a *subspace* is a smaller vector space that is contained in the original vector space. For example, in 3-dimensional space, the x - y plane is a 2-dimensional subspace because any linear combination of vectors in that plane remain in the plane: thus it is a vector space in its own right. Subspaces are often constructed as a span (linear combination) of some set of vectors, e.g. the span of $\{[-2, 0, 0], [0, 5, 0]\}$ gives the x - y plane in 3D space. In infinite dimensions subspaces are defined in exactly the same way.

Although they sound similar, you should not confuse a *subspace* with a *subset*. Any set of vectors can be a subset of the original vector space. A subspace, however, has to still itself be a vector space in its own right.

Example B.3. Consider the subset of $\mathcal{L}_2(P)$ comprised of all functions that have zero mean under $\mathbb{P}$, i.e. functions f where $\mathbb{E}[f(Z)] = 0$. This subset $\mathcal{V}$ is actually a *subspace* because it is a vector space. To verify that we just need to check that $\mathcal{V}$ is a vector space: if we add and scale elements with it, the results are still in $\mathcal{V}$.

To that end, let f and g be any functions in $\mathcal{V}$ and let c be a scalar constant. Since

$$\mathbb{E}[cf(Z) + g(Z)] = c\mathbb{E}[f(Z)] + \mathbb{E}[g(Z)] = 0,$$

$cf + g$ satisfies the condition required for membership in $\mathcal{V}$, proving it is a vector space.

Exercise B.6. Let $\mathbb{P}$ be the standard uniform distribution. Convince yourself that in $\mathcal{L}_2(\mathbb{P})$, the set of linear functions $\{f : f(z) = mz + b\}$ is a subspace and is spanned by $\{f(z) = 1, f(z) = z\}$ (thus making it a two-dimensional subspace!). Can you think of a subspace of this subspace? Can you think of a subset of linear functions that is *not* a subspace?

Now that we have our definition of subspaces we can define orthogonal projection in the context of inner product spaces.

Orthogonal Projection

Definition B.4. The orthogonal *projection* of a vector v onto a (closed) subspace $\mathcal{T}$ is defined to be the vector in $\mathcal{T}$ that is closest to v . In this book we'll use the notation

$$\mathcal{T}v = \operatorname{argmin}_{t \in \mathcal{T}} \|v - t\|$$

to denote orthogonal projections.

Does Definition B.4 make sense?

For the definition above to make sense, we have to prove that there *is* a minimum-distance point to v in $\mathcal{T}$: it could be that there is none (what is the one rational number that is closest to π ?) or that it is not unique (what is the integer that's closest to $1/2$?). For the purposes of this book you can take it as fact that it is well-defined. The proof is not complicated, but it requires more fluency with convergence and infimums than I'd like to assume you have (as well as the parallelogram law). A key required assumption which is satisfied in all cases within this book is that the subspace is *closed*, i.e. that it contains its limits. If you are curious, you can take a screenshot of this definition and ask a large language model to supply the proof.

In terms of intuition, you can think of projection as “dropping down” the vector v onto the “plane” $\mathcal{T}$. The idea is that the projection $\mathcal{T}v$ is the “best approximation” of v that we can manage in $\mathcal{T}$.

In the case that $\mathcal{T}$ is a one-dimensional subspace of the larger vector space $\mathcal{V}$ we have a handy formula for computing the projection.

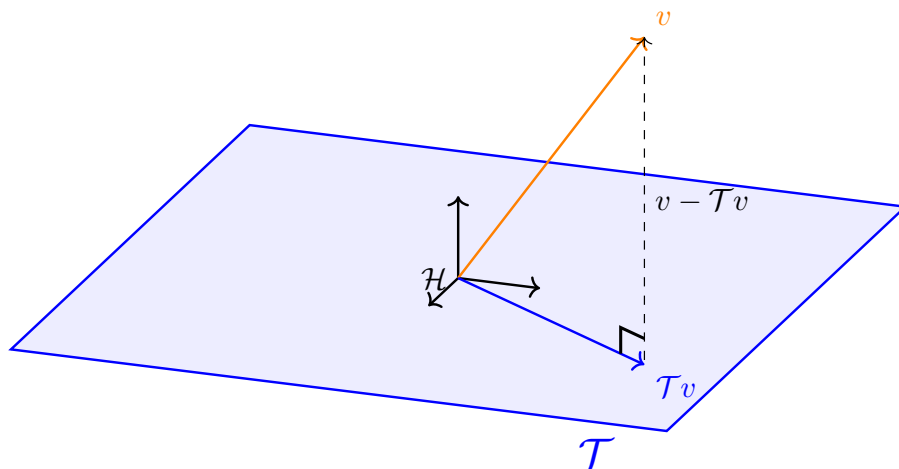


Figure B.2: A visual representation of orthogonal projection. Notice that the residual $v - \mathcal{T}v$ is orthogonal to all vectors in $\mathcal{T}$.

1D projection

Theorem B.2. If $\mathcal{T} = \{c\tilde{t} : c \in \mathbb{R}\}$ for some fixed, nonzero vector $\tilde{t}$, then

$$\mathcal{T}v = \left\langle v, \frac{t}{\|t\|} \right\rangle \frac{t}{\|t\|}$$

for any $t \in \mathcal{T}$.

Proof of Theorem B.2

The projection $\mathcal{T}v$ is defined to be the vector t^* in $\mathcal{T}$ that minimizes the distance $\|t - v\|$. Pick any nonzero vector $t \in \mathcal{T}$. Since every vector in $\mathcal{T}$ can be written as ct for some scalar c , we can write the minimizer $t^* = c^*t$. Using this and the fact that minimizing the distance is the same as minimizing the squared distance, the optimization problem is to find c that minimizes

$$\begin{aligned} \|ct - v\|^2 &= \langle ct - v, ct - v \rangle \\ &= c^2 \langle t, t \rangle - 2c \langle t, v \rangle + \langle v, v \rangle \end{aligned}$$

To find the minimizer we can take the derivative of this with respect to c and set it equal to zero, giving $2\|t\|^2 c^* - 2\langle t, v \rangle = 0$. Solving for c^* and plugging into $t^* = c^*t$ gives the result.

This formula gives us a nice geometric interpretation to the inner product if you consider a unit vector $\tilde{t}$. In this case, the projection formula simplifies to $\mathcal{T}v = \langle v, \tilde{t} \rangle \tilde{t}$. The inner product

of v with the unit vector $\tilde{t}$ is therefore the length of the projection of f onto the span of $\tilde{t}$. The inner product with vectors t of different norm will give the same number, just scaled by that norm. Formally, if we let $\mathcal{T}$ be the span of t ,

$$\langle v, t \rangle = \|\mathcal{T}v\| \|t\|.$$

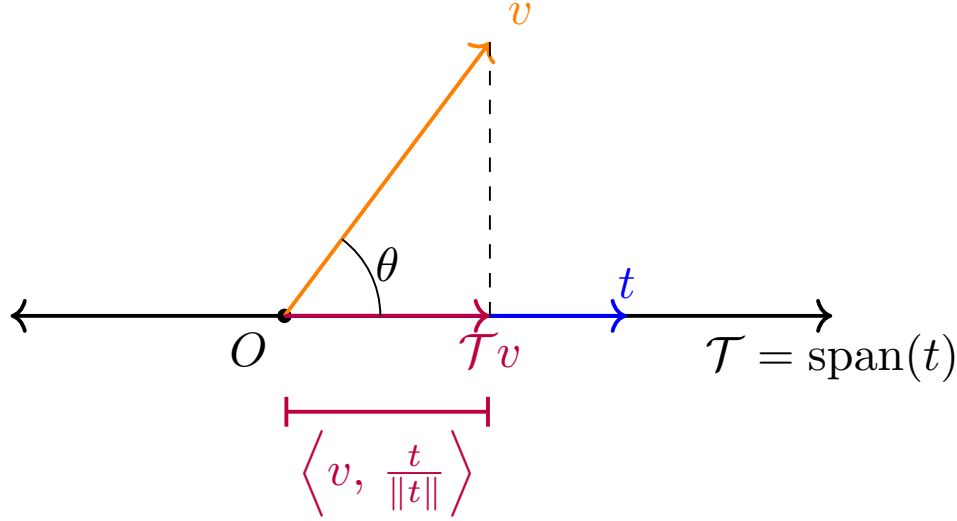


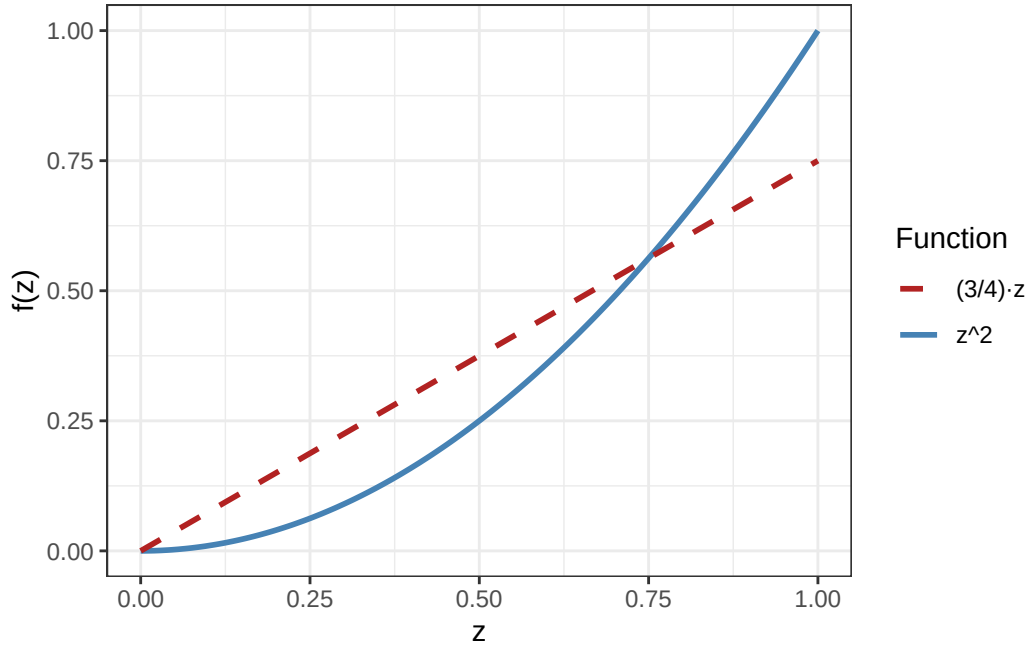
Figure B.3: Visual representation of the inner product giving the length of a projection.

From this picture you can see that the angle between any $t \in \mathcal{T}$ (not necessarily norm 1) and v can be extracted by taking the inverse cosine of the lengths of the “adjacent” over the “hypotenuse” (the classical definition of cosine). Those lengths are $\langle v, t/\|t\| \rangle$ and $\|v\|$. Dividing and pulling out the norm gives the law of cosines $\cos(\theta) = \frac{\langle v, t \rangle}{\|v\| \|t\|}$.

Example B.4. Let’s presume that $\mathbb{P}(Z)$ is the uniform distribution over $[0, 1]$ and let $\mathcal{T}$ be the subspace of $\mathcal{L}_2(\mathbb{P})$ comprising of all linear functions with no intercept, i.e. functions that can be written as $f(z) = cz$ for some slope c . You can verify that this is a subspace.

Let’s let $v(z) = z^2$ and figure out what the projection of this function is onto $\mathcal{T}$. In other words, what is the linear function that best approximates z^2 in a mean-squared sense over the domain $[0, 1]$? To do this we’ll use our formula: $\mathcal{T}v = \frac{\langle t, v \rangle}{\|t\|^2} t$

Pick $t(z) = z$ as the representative from $\mathcal{T}$. Then $\langle t, v \rangle = \int_0^1 z z^2 dz = 1/4$ and $\|t\|^2 = \int_0^1 z^2 dz = 1/3$. Thus $\mathcal{T}v = \frac{3}{4}x$. Visually your eye might agree that this is the best (no-intercept) linear approximation:



We call what's left over when you subtract the projection from the original vector the *residual*. An important geometric property of projections that you may recall from linear algebra (and which I have already been showing in the figures) is that residuals are always orthogonal to the entire subspace that gets projected on (see Figure B.2).

Orthogonality of Residuals

Theorem B.3. *For any subspace $\mathcal{T} \subseteq \mathcal{V}$ and vector $v \in \mathcal{V}$ in a Hilbert space for which $\mathcal{T}v$ exists and is unique, we have that $t \perp (v - \mathcal{T}v)$ for all $t \in \mathcal{T}$.*

Proof of Theorem B.3

For any $t \in \mathcal{T}$ we compute the (squared) distance between v and a perturbed version of its projection $\mathcal{T}(v) + \epsilon t$ for an arbitrary nonzero perturbation direction $t \in \mathcal{T}$:

$$\begin{aligned}
 d_t(\epsilon) &= \|v - (\mathcal{T}v + \epsilon t)\|^2 \\
 &= \langle v - \mathcal{T}v + \epsilon t, v - \mathcal{T}v + \epsilon t \rangle \\
 &= \|v - \mathcal{T}v\|^2 + 2\epsilon \langle v - \mathcal{T}v, t \rangle + \epsilon^2 \|t\|^2
 \end{aligned}$$

The fact that $\mathcal{T}v$ is the minimizer of the distance to v means that if we perturb it by ϵ in any direction t , we must be increasing the distance. Therefore for all t , $d_t(\epsilon)$ is minimized at $\epsilon = 0$. Since d_t is a smooth function in ϵ for any t (a quadratic, in fact), if we take the

derivative at $\epsilon = 0$ it must be zero:

$$0 = d'_t(0) = \langle v - \mathcal{T}v, t \rangle + 0 \cdot \|t\|^2.$$

which establishes the orthogonality $\langle v - \mathcal{T}v, t \rangle = 0$ for all $t \in \mathcal{T}$.

A special case of this is when $t = \mathcal{T}v$, which by definition is a vector in $\mathcal{T}$. The residual and the projection form two sides of a right triangle with the original vector as the hypotenuse. Therefore we often encounter the pythagorean theorem in the context of projection:

$$\|v\|^2 = \|\mathcal{T}v\|^2 + \|v - \mathcal{T}v\|^2.$$

Exercise B.7. Let $\mathbb{P}$ be the standard uniform distribution. Your friend claims that the $\mathcal{L}_2(\mathbb{P})$ projection of the function $f(z) = 1$ onto the subspace of all polynomials has norm equal to $1/2$. Without any calculation, explain why this is impossible. Again without any calculation, explain what the norm of the desired projection actually is.

At this point you should recognize we're abusing notation by using $\mathcal{T}$ as both a subspace and as a projection operator (function) that takes a given vector v and returns another vector which is the projection onto $\mathcal{T}$. Besides convenience, one reason to write it like this is that $\mathcal{T}v$ evokes the matrix-vector multiplication notation Ax . In fact, orthogonal projection *is* synonymous with a special kind of matrix multiplication in $\mathbb{R}^d$. More generally, you should bring that intuition with you: projection is a special kind of linear mapping.

Linearity of Orthogonal Projection

Theorem B.4. *For any subspace $\mathcal{T}$, projection is a linear mapping in the sense that $\mathcal{T}(cv + u) = c\mathcal{T}v + \mathcal{T}u$.*

Proof of Theorem B.4

Fix $v, u \in V$ and $c \in \mathbb{R}$. For any vector v we know we can project it onto some space $\mathcal{T}$ to obtain the projection $\mathcal{T}v \in \mathcal{T}$ and a residual $v - \mathcal{T}v$ orthogonal to $\mathcal{T}$. Therefore let's decompose

$$\begin{aligned} cv + u &= \mathcal{T}(cv + u) + [cv + u - \mathcal{T}(cv + u)] \\ cv &= c(\mathcal{T}v + [v - \mathcal{T}v]) \\ u &= \mathcal{T}v + [u - \mathcal{T}v] \end{aligned}$$

The two equations on the bottom sum to the one on the top. The first term on the right-hand side of each equation is a vector in $\mathcal{T}$ and the second is a vector orthogonal to it. The part of $cv + u$ that is in $\mathcal{T}$ can be composed only of the parts of v and u that are in $\mathcal{T}$ because no vectors that are orthogonal to a subspace can ever sum to a nonzero

vector within the subspace. Thus $\mathcal{T}(cv + u) = c\mathcal{T}v + \mathcal{T}u$ which proves the linearity.

The proof of Theorem B.4 is based on the common theme that many interesting vector spaces can be built up as a kind of “sum” of subspaces.

Orthogonal Sum and Complement

Definition B.5. Two subspaces $\mathcal{V}_1$ and $\mathcal{V}_2$ are orthogonal if every vector $v_1 \in \mathcal{V}_1$ is orthogonal to every vector $v_2 \in \mathcal{V}_2$. A vector space $\mathcal{V}$ is the orthogonal sum of subspaces $\mathcal{V}_1$ and $\mathcal{V}_2$ if every vector $v \in \mathcal{V}$ can be written as $v = v_1 + v_2$ for some (unique) vectors $v_1 \in \mathcal{V}_1$ and $v_2 \in \mathcal{V}_2$. If this is the case we write $\mathcal{V} = \mathcal{V}_1 \oplus \mathcal{V}_2$. When two orthogonal subspaces sum to the total vector space $\mathcal{V}$, we say that they are orthogonal complements of each other and usually write them as $\mathcal{T}$ and $\mathcal{T}^\perp$ (which is which does not matter).

When I think of orthogonal sums, I usually think of 3D space being the “sum” of the x - y plane and the z -axis. The x - y plane is itself the sum of the x - and y -axes. Alternatively you could formulate 3D space as the sum of the z - y plane and the x -axis. This shows that there are many ways to “break up” a space into orthogonal subspaces. The “named” axes don’t matter either, I could also break up 2D space into the sum $\mathbb{R}^2 = \{c[1, 1] : c \in \mathbb{R}\} \oplus \{c[1, -1] : c \in \mathbb{R}\}$.

In the proof of the linearity of orthogonal projection, we used the fact that by the orthogonality of projection, $\mathcal{V} = \mathcal{T} \oplus \mathcal{T}^\perp$. We broke the original vectors and their sum into their projections and residuals and then noted that by orthogonality the only way to make the sums match up was to match up the parts corresponding to each orthogonal complement.

This also gives a general method to decompose a vector space into an orthogonal sum. First pick an arbitrary nonzero vector t . Then project every other vector onto its span $\mathcal{T}$ and take all the residuals. The set of projections and set of residuals are then orthogonal subspaces that sum to the full vector space.

Exercise B.8. Prove the above claim by following these steps:

1. Show that the set of projections $\{\mathcal{T}v : v \in \mathcal{V}\}$ and the set of residuals $\{v - \mathcal{T}v : v \in \mathcal{V}\}$ are indeed vector spaces (check that the sum of two vectors that satisfy the definition again satisfies the definition...)
2. Prove that any vector from the projection subspace is orthogonal to any vector from the residual subspace (already done)
3. Find the unique way to write every $v \in \mathcal{V}$ as a sum of two vectors from these subspaces (practically already done)

Example B.5. Let $\mathbb{P}$ be the standard uniform distribution and let $\mathcal{V} = \{f : f(z) = mz + b\}$ be the subspace of linear functions in $\mathcal{L}_2(\mathbb{P})$. Let’s find a way to decompose this subspace into an orthogonal sum.

Pick $f(z) = 1$. The span of f is clearly $\mathcal{T} = \{z \mapsto c : c \in \mathbb{R}\}$. Using our 1D projection formula and taking the representative $f(z) = 1$ (which has norm 1), the residual of any function $g(z) = mz + b$ projected onto $\mathcal{T}$ is given by

$$\begin{aligned} (mz + b) - \mathcal{T}(mz + b) &= (mz + b) - \int_0^1 mz + b \, dz \\ &= m(z - 1/2) \end{aligned}$$

Therefore

$$\{z \mapsto mz + b : m, b \in \mathbb{R}\} = \{z \mapsto c : c \in \mathbb{R}\} \oplus \{z \mapsto m(z - 1/2) : m \in \mathbb{R}\}$$

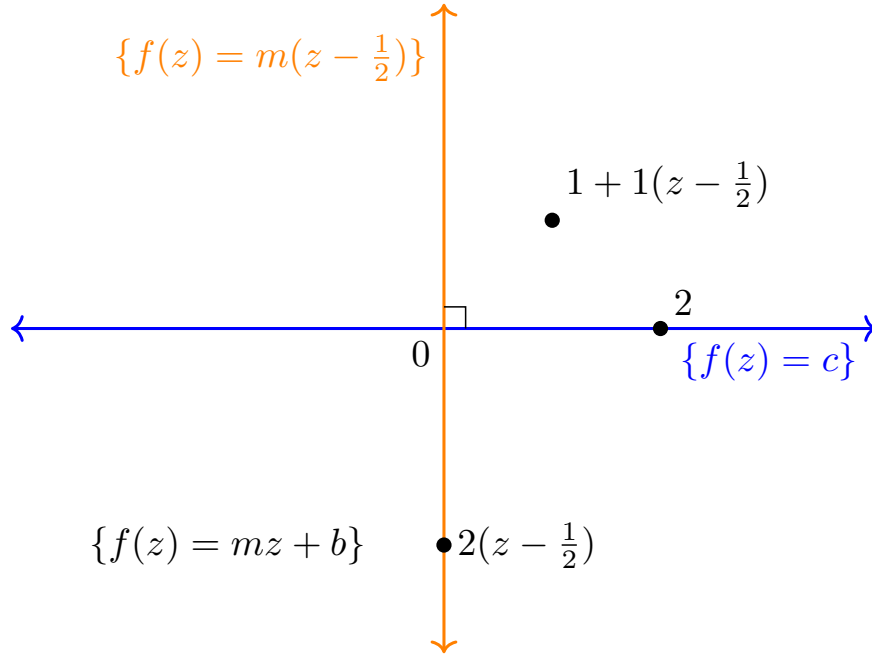


Figure B.4: A visual representation of the orthogonal sum. Each “point” in this vector space is a linear function defined over $z \in [0, 1]$. The two orthogonal “axes” are the constant functions and the functions that have the form $m(z - \frac{1}{2})$. If you picked a different initial axis, you could construct a different orthogonal “coordinate system” for these functions.

B.3.3 Cauchy-Schwarz

One important consequence of the general pythagorean theorem is that we can easily bound inner products with a product of norms

Cauchy-Schwarz Inequality

Theorem B.5.

$$|\langle f, g \rangle| \leq \|f\| \|g\|$$

for any two elements f, g in the space. The equality is obtained if and only if $f = cg$ for some scalar c .

Proof of Theorem B.5

Let $\mathcal{G}$ be the span of g and consider the projection $\mathcal{G}f = \frac{\langle g, f \rangle}{\|g\|^2} g$. Recall that we previously used our one-dimensional projection formula to show

$$\langle f, g \rangle = \|\mathcal{G}f\| \|g\|,$$

a geometric interpretation to the inner product as the scaled length of the “adjacent” side of a right triangle. The Cauchy-Schwarz inequality is immediate from this because $\|\mathcal{G}f\| \leq \|f\|$ since these are the side and hypotenuse of a right triangle, respectively. Equality holds if and only if $f \in \mathcal{G}$ already, i.e. if $f = cg$ for some scalar c .

One way to remember the Cauchy-Schwarz inequality is that it makes it sensible to define an “angle” as $\cos(\theta) = \frac{\langle f, g \rangle}{\|f\| \|g\|}$ since the right-hand side is guaranteed to be between 1 and -1. The probabilistic version of this in $\mathcal{L}_2(\mathbb{P})$ is that for two zero-mean functions $\mathbb{E}[f(Z)] = \mathbb{E}[g(Z)] = 0$,

$$\frac{\langle f, g \rangle}{\|f\| \|g\|} = \frac{\mathbb{C}[f(Z), g(Z)]}{\sqrt{\mathbb{V}[f] \mathbb{V}[g]}},$$

which is the correlation between the random variables $f(Z)$ and $g(Z)$. This must also be between -1 and 1. This also gives a geometric interpretation to the correlation coefficient: it’s a measure of the “angle” between two random variables viewed as vectors in $\mathcal{L}_2(\mathbb{P})$.

Example B.6 (Bounding an Integral). Suppose I know $f \in \mathcal{L}_2(\mathbb{P})$. In many problems we might end up with an expression like $\mathbb{E}[f(Z)]$ which we might not know how to compute without knowing exactly what f is. However, Cauchy-Schwarz gives us one useful way to approximate such a term in the sense that we can at least say it’s no bigger than a certain number. To wit,

$$|\mathbb{E}[f(Z)]| = \left| \int f(Z) \cdot 1 \, d\mathbb{P} \right| = |\langle f, 1 \rangle| \leq \|f\|.$$

We even know that the bound is *sharp* (we can’t do any better without knowing more). That’s because if $f = c$ (a constant), then $\mathbb{E}[c] = \|c\|$.

B.3.4 Triangle Inequality

Lastly, we have the aptly-named triangle-inequality, which essentially says that the fastest way from point A to point B is a straight line. If you take a detour to point C, the total distance that you travel is larger than if you had gone directly to B.

Triangle Inequality

Theorem B.6. For any norm $\|\cdot\|$ and vectors a, b, c ,

$$\|a - b\| \leq \|a - c\| + \|c - b\|.$$

Proof of Theorem B.6

Let $u = a - c$ and $v = c - b$ so that $a - b = u + v$.

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \\ &\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 \quad (\text{Cauchy-Schwarz}) \\ &= (\|u\| + \|v\|)^2\end{aligned}$$

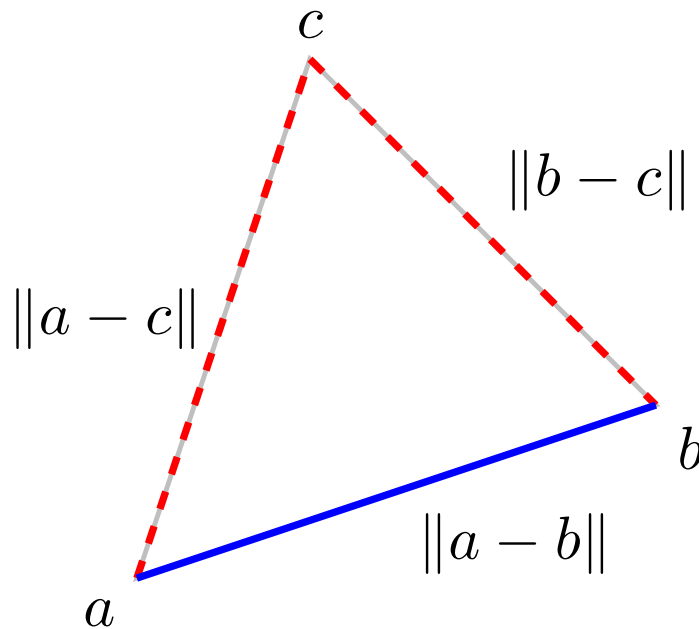


Figure B.5: Illustration of the triangle inequality.

All norms that are induced by inner products satisfy the triangle inequality by design, as we

have proven. However, the triangle inequality is actually part of the definition of a norm: any function that someone wants to call a norm must satisfy the triangle inequality in order for them to call it that.

Sometimes we are interested in proving that one side of a triangle is small, but what we know about are the other two sides. That's when the triangle inequality comes in handy.

Example B.7. If I have a sequence of real numbers $x_n \rightarrow x$ and another $y_n \rightarrow y$, is it true that $x_n y_n \rightarrow xy$?

Warning

If you don't understand (in a precise way) what I mean by convergence of a sequence you should review Section A.1.

Let's consider the difference $|x_n y_n - xy|$. To prove the claim we have to show that this is eventually always smaller than any given ϵ . We can do this by taking a "detour" to the point(s) $x_n y$:

$$|x_n y_n - xy| \leq |x_n y_n - x_n y| + |x_n y - xy|.$$

Perhaps it goes without saying, but the absolute value is a norm on $\mathbb{R}$ induced by the inner product $\langle x, y \rangle = xy$ so we are justified in applying the triangle inequality. Our purpose in taking this detour is that we're trying to somehow reduce the distance $|x_n y_n - xy|$ into some combination of $|x_n - x|$ and $|y_n - y|$ because we know we can make those distances as small as we like by taking n large enough (that's the definition of convergence).

For each of the terms above, we can now use Cauchy-Schwarz, e.g. $|x_n y_n - x_n y| = |x_n(y_n - y)| \leq |x_n||y_n - y|$, therefore

$$|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x|.$$

We're almost there but we have the term $|x_n|$ which is neither a constant like $|y|$ nor converging to zero. Here again we can use the triangle inequality on $|x_n|$ and take a detour to x to expand $|x_n| = |x_n - 0| = |x_n - x| + |x|$, leaving us with

$$|x_n y_n - xy| \leq |x||y_n - y| + |x_n - x||y_n - y| + |y||x_n - x|.$$

Now everything is in terms of constants and the two distances which we know go to zero. Therefore we have the desired convergence.

When you first see proofs like the one in the example above it can feel overwhelming: how did I know where to use the triangle inequality? How did I know when to use Cauchy-Schwarz? It all seems like black magic and cheap tricks that you can understand in retrospect, but could

never have constructed yourself. But there actually are certain “tells” that you can use to build your intuition.

For example, when you see a norm of a sequence, you should immediately want to approximate it by the norm of the limit $|x_n| \approx |x|$ for convenience since the latter is just a constant that you can deal with easily. The triangle inequality then comes to mind as a way to do this since we can “detour” x_n to x and then get the approximation $|x_n| \leq |x_n - x| + |x|$. Cauchy-Schwarz should come to mind any time you see pretty much any inner product. $|yx_n - yx|$ is a mess but $|y||x_n - x|$ is something we can understand. With continued practice you will get very good at just “seeing” what to do. If you’re interested in that I recommend picking up a real analysis textbook and working through the problems.

B.4 Riesz Representation

In this section we’ll talk about *Riesz representation*. The basic idea here is that any linear function whose inputs are elements of a vector space can be entirely characterized by its direction of steepest ascent and by its slope in that direction. We’ll first walk through the ideas in finite-dimensional inner product spaces where things are easier to visualize and we don’t need any definitions from real analysis before proceeding to the general situation in possibly infinite-dimensional Hilbert spaces.

B.4.1 Warm-Up in $\mathbb{R}^d$

You may recall from linear algebra or vector calculus that many useful functions map vectors to other vectors or vectors to numbers. In $\mathbb{R}^d$, given a $p \times d$ matrix A , the matrix-vector “multiplication” Av can actually be seen as a function that maps $v \in \mathbb{R}^d$ to $Av \in \mathbb{R}^p$. Similarly, the vector dot product $v \mapsto a \cdot v$ for some vectors $a, v \in \mathbb{R}^d$ maps $v \in \mathbb{R}^d$ to $a \cdot v \in \mathbb{R}$. Both of these example functions on vectors are *linear* in the sense that $f(cv + u) = cf(v) + f(u)$ for a scalar c and vectors v, u .

It’s clear that the dot product between an argument v and some fixed vector a is linear in the argument v , but actually the relationship goes both ways: *all* linear functions can be written as an inner (dot) product between the argument and some other fixed vector in the vector space. Formally, for any function f which is linear and maps $\mathbb{R}^d$ to $\mathbb{R}$, there is some fixed $t_f \in \mathbb{R}^d$ that “represents” the function in the sense that for all $f(v) = \langle t_f, v \rangle$ for all $v \in \mathbb{R}^d$. We call the vector t_f the “Riesz representer” of the function f .

Mechanically this is pretty easy to show. To start, we can always write v as a sum of its components along the orthonormal basis vectors e_j that span our vector space: $v = \sum_j v_j e_j$.

Now we apply the function f to both sides and use the linearity:

$$f(v) = f\left(\sum_j v_j e_j\right) = \sum_j v_j f(e_j) = \langle v, t_f \rangle$$

where we've defined the vector $t_f = [f(e_1), f(e_2), \dots, f(e_d)]$. Since f maps vectors to numbers, each element $f(e_j)$ is indeed a scalar value. It's important to recognize that t_f is a fixed vector associated with the fixed function f : it doesn't depend on the argument v .

While the proof is simple, it doesn't give much intuition for what the result *means* or give any sense of *why* it's true. To do that it's helpful to look at some pictures. In the figure below I've drawn the graph of some linear function of vectors in $\mathbb{R}^2$: it looks like a plane that passes through the origin. For each vector $v = [v_1, v_2]$ on the $\mathbb{R}^2$ plane, you can find the scalar value of $f(v)$ by drawing a vertical up to the graph that represents f .

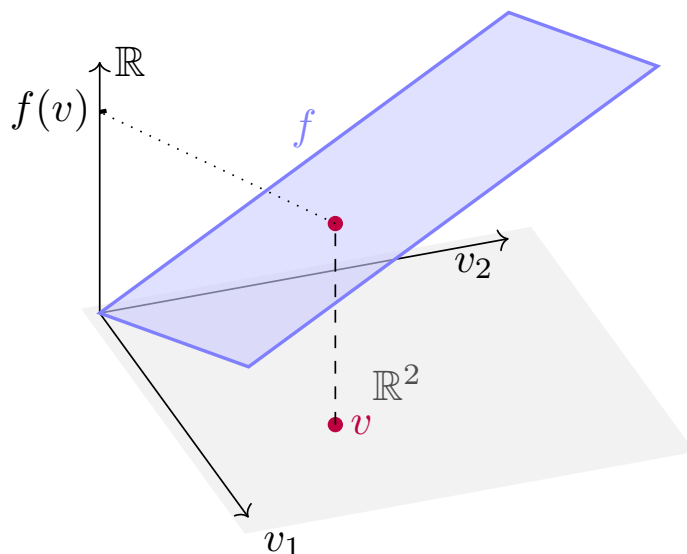


Figure B.6: A linear function mapping $v \in \mathbb{R}^2$ to $f(v) \in \mathbb{R}$.

It should be somewhat intuitive to you that a linear function has a linear (planar) graph but nonetheless let's think about why that is. The graph must pass through the origin because by linearity $f(0) = 0 \times f(v) = 0$. Now, for any fixed vector v , the graph of ϵ vs $f(\epsilon v) = \epsilon f(v)$ is a straight line between with intercept 0 and slope $f(v)$. Thus the graph of f is linear in all directions and therefore planar.

Let's look at this graph from the top down. I've drawn this in the figure and you can see that the level sets of f (the "elevation lines" on a map) are straight lines that all go in the same direction. The direction orthogonal to these level sets is the direction in which f is increasing the fastest. If you stood on the surface, that would be the "uphill" direction. I've also drawn

a similar picture for another linear function that I call g . For g you can see that the uphill direction is different and that the level sets are closer together, meaning that the climb is “steeper” in the uphill direction than it is for f .

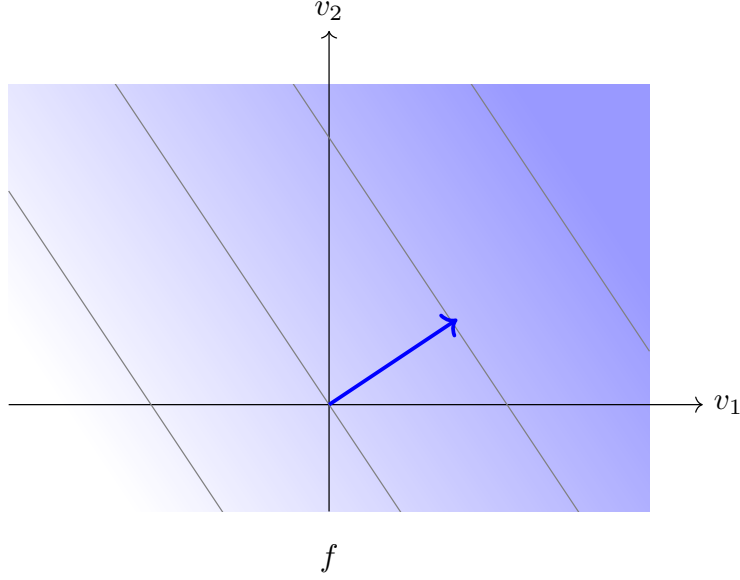


Figure B.7: The same linear function as seen in Figure B.6, but seen from the top down. Darker blue indicates larger values of $f(v)$. Grey lines are the level sets $f(v) = c$, spaced at equal intervals in c so that the density of those lines indicates the slope.

Looking at the two examples, you may realize that the uphill direction and the steepness in that direction are *everything* you need to know in order to characterize any linear function f . The uphill direction can be represented by a unit vector and the steepness is a factor that it is scaled by. We call that scaled unit vector t_f , our vector representer of f . This vector contains all the information there is to know about f , and vice versa. Technically they are not the same object, but we can always reconstruct one from the other.

It remains to be shown that this vector is the same representer t_f that we derived above. To find the “most uphill” direction, we see how much we climb if we take a step v of unit length $\|v\| = 1$ in any direction away from some starting point. It doesn’t matter from where we start so for convenience we’ll start from zero. Formally, the “most uphill” direction is therefore

$$\begin{aligned} v^* &= \operatorname{argmax}_{v \in \mathbb{R}^d: \|v\|=1} f(v) \\ &= \operatorname{argmax}_{v \in \mathbb{R}^d: \|v\|=1} \sum v_i f(e_i) \quad (\text{linearity}) \end{aligned}$$

i.e. we want to maximize the inner product $\langle v, t_f \rangle$ over all v with norm 1. By Cauchy-Schwarz, this is attained by letting $v = t_f / \|t_f\|$. If we look at the amount by which f climbs over one

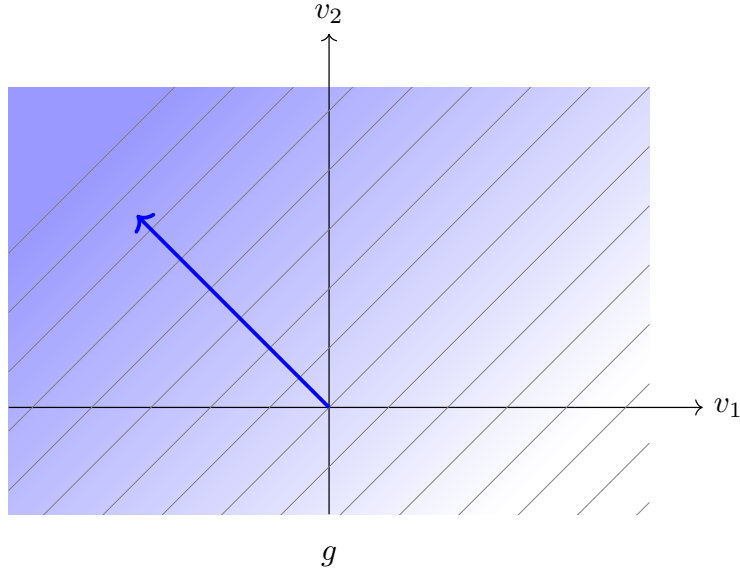


Figure B.8: A different linear function from the top down.

unit step in this direction, it is $f(t_f/\|t_f\|) = \langle t_f, t_f \rangle / \|t_f\| = \|t_f\|$. Therefore the representer t_f is nothing more than the direction in which f is increasing the most, with its norm indicating how quickly f increases in that direction.

Exercise B.9. How is the Riesz representer of a linear function from $\mathbb{R}^d$ to $\mathbb{R}$ related to its gradient?

I've explained intuitively why this representer vector uniquely identifies any linear function, but I haven't explained why taking the inner product of the argument with the representer is equivalent to evaluating the function. To understand that again I think it helps to look at the pictures above that show the level sets of the function. If you imagine walking along one of those level sets (orthogonal to the representer), the value of the function does not change. In other words, the function f is completely insensitive to any motion at any point that goes in a direction that is orthogonal to the direction of steepest increase. Let's decompose $\mathcal{V}$ orthogonally into $\mathcal{T} = \text{span}(t_f)$ and $\mathcal{T}^\perp$. What I've said is formally that $f(v + t^\perp) = f(v)$ for any motion $t^\perp$ in $\mathcal{T}^\perp$.

Now imagine moving in an arbitrary direction u . Whatever direction you go, you can decompose it into a part that is in the direction of steepest ascent and the part that is in the orthogonal complement of that subspace, i.e. write $u = t + t^\perp$. By the linearity of f , $f(u) = f(t) + f(t^\perp) = f(t)$. In other words, f doesn't even care about any *component* of motion that isn't in the direction of steepest ascent. Finally, let's think about what the vector t is: it's the part of u that lies in $\mathcal{T}$, i.e. the projection $\mathcal{T}u = \left\langle u, \frac{t_f}{\|t_f\|} \right\rangle \frac{t_f}{\|t_f\|}$. The inner product

is just a constant so when we evaluate $f(u) = f(\mathcal{T}u)$ it gets passed right through and we're left with

$$f(u) = f(\mathcal{T}u) = \left\langle u, \frac{t_f}{\|t_f\|} \right\rangle f\left(\frac{t_f}{\|t_f\|}\right) = \left\langle u, \frac{t_f}{\|t_f\|} \right\rangle \|t_f\|$$

where the last equality obtains because the increase in f under one unit step in the direction of maximal increase is $\|t_f\|$, as we just showed. Therefore $f(u) = \langle u, t_f \rangle$ as desired.

The overall lesson is that linear functions completely ignore motion in any direction that isn't the direction of steepest increase. Because of this, the only part of the input that matters is the part in the direction of steepest increase, which is given by projecting the input onto that direction. Since the projection is related to the inner product, we can replace function evaluation with taking that inner product as long as we choose a representative in the subspace of maximum increase that has a norm equivalent to the increase in f when moving one unit in that direction.

To be clear, none of this reasoning proves the fact that every linear function on $\mathbb{R}^d$ has a Riesz representer any better than the 2-line argument I gave at the beginning of this section. But I hope that walking through this has helped you “feel it” in your bones rather than “know it” as an abstract fact.

B.4.2 Infinite-Dimensional Riesz Representation

Now we'll talk about the more general case when the Hilbert space of interest could be infinite-dimensional. For the most part we'll be interested in $\mathcal{L}_2$ or subspaces thereof. If we talk about functions defined *on* $\mathcal{L}_2$, we're talking about functions *of functions*. In functional analysis, people sometimes call these *operators*, but there's really nothing special about them: they are still functions. When an operator has real-valued output (i.e. maps functions to numbers), it's often called a *functional*.

Example B.8 (Expectation Functional). The function $f \mapsto \mathbb{E}_{\mathbb{P}}[f(Z)]$ is a functional (a linear functional, in fact). The expectation functional is defined on all of $\mathcal{L}_2(\mathbb{P})$ since every square-integrable function is integrable (see Example B.6).

In general Hilbert spaces the story of Riesz representation is essentially the same except there is one little wrinkle: not all linear functions actually have a direction of steepest ascent. The problem is that there are too many “directions” to go in within infinite-dimensional space. No matter what direction you “look” (e.g. imagine standing at the origin), there may be some other direction you could look in where the slope (the unit rise in the function over a unit change in the input) is steeper! In contrast, in 3D (or d -dimensional) space, the fact that there are only 3 (d) dimensions means that there is *always* some direction of steepest ascent: you quickly run out of directions to look for steeper slopes.

Example B.9 (Evaluation at a Point). Let $Z \sim \text{Unif}(-1, 1)$ and let $T : f \mapsto f(0)$ be the functional that takes $f \in \mathcal{L}_2(\mathbb{P})$, evaluates it at the point $z = 0$, and returns the result. Clearly T is linear under the definition of function addition and scaling in $\mathcal{L}_2$ because

$$T(cf + g) = (cf + g)(0) = cf(0) + g(0) = cTf + Tg.$$

Interestingly, however, T does not have a “direction of steepest ascent”. To understand this, let’s first define the slope of T as the rise in T over a unit change in the input f . Equivalently, we may consider the rise in T divided by the amount of change in the input:

$$\frac{T(f) - T(f + h)}{\|h\|}.$$

Since T is linear, the slope in a given direction is the same at all points so we can take $f = 0$ to simplify things without loss of generality. Now let’s pick some directions h (these are functions; recall functions are the vectors in $\mathcal{L}_2$) and see what the magnitude of the slope $\frac{|Th|}{\|h\|}$ is. Consider the functions (directions, vectors)

$$h_n(z) = 2\sqrt{n}1\left(-\frac{1}{n} \leq z \leq \frac{1}{n}\right).$$

These functions have norms

$$\|h_n\|^2 = \frac{1}{2} \int_{-n^{-1}}^{n^{-1}} 2^2 \sqrt{n}^2 \, dz = 1.$$

On the other hand,

$$Th_n = h_n(0) = 2\sqrt{n} \rightarrow \infty.$$

By increasing $n \rightarrow \infty$ I can find a direction $h = h_n$ where the slope $Th/\|h\| = 2\sqrt{n}$ is bigger than any given number. It is therefore not possible to find a direction of steepest ascent: there is always a steeper slope in another direction!

In the field of functional analysis, functionals that don’t have a direction of steepest ascent are called *unbounded*.

Bounded Linear Functional

Definition B.6. A linear functional $T : \mathcal{V} \rightarrow \mathbb{R}$ for a Hilbert space $\mathcal{V}$ is called bounded if

$$\sup_{v \in \mathcal{V}} \frac{|Tv|}{\|v\|} < \infty.$$

The precise value of the supremum is called the *operator norm* of the functional, sometimes written $\|T\|_{\text{op}}$.

“Boundedness” in this sense is not the same as usual sense of the word, which would be $\sup_v \|Tv\| < \infty$. No linear function can be bounded in the traditional sense because no matter what Tv is for some vector v , I can scale v by a large constant c and make $T(cv) = cTv$ as large as I want. But that’s not what boundedness means in the sense of functional analysis. In a functional analysis context, boundedness means that there is a maximum value to the *slope* of T , not that the *output* of T has a maximum value.

Exercise B.10. Every functional of the form $v \mapsto \langle t, v \rangle$ is linear by the definition of any inner product. Use the Cauchy-Schwarz inequality to prove that any such functional is also bounded.

Now we can finally state the general-purpose Riesz representation theorem:

Riesz Representation

Theorem B.7. *Every bounded linear functional $T : \mathcal{V} \rightarrow \mathbb{R}$ on a Hilbert space can be written as*

$$Tv = \langle t, v \rangle$$

for some fixed $t \in \mathcal{V}$ which we call the Riesz representer of T .

Proof of Theorem B.7

First we’ll quickly prove that T is continuous. It’s counterintuitive, but linear functionals in infinite dimensions can be discontinuous. In fact, continuity is the same thing as boundedness, but we’ll just prove the implication in one direction. Consider a sequence of vectors $v_n \rightarrow v$ in $\mathcal{V}$. Since T is bounded, $|Tv_n - Tv| = |T(v_n - v)| \leq \|T\|_{\text{op}} \|v_n - v\| \rightarrow 0$, proving that T is continuous (a sequence of inputs converging to a limit means the sequence of outputs converges to the function evaluated at that limit).

Based on our intuition from the finite dimensional case, let’s take $\tilde{t}$ to be the maximizer of Tv over all unit-norm vectors v . Since T is bounded, $|Tv|$ has a maximum value over the set of unit-norm vectors. Moreover, since the set of unit-norm vectors is closed and bounded and T is continuous, T attains its supremum somewhere: there is a unit-norm $\tilde{t}$ such that $T\tilde{t} = \sup_{\|v\|=1} Tv$ (this is called the extreme value theorem; you can look it up but it’s not important to get the gist of things).

Now we’ll show that the function T only responds to changes in its input that are aligned with $\tilde{t}$, just like in the finite-dimensional case. Let $\mathcal{T}$ be the span of $\tilde{t}$. Pick an arbitrary input v and consider the decomposition

$$v = \underbrace{(v - \mathcal{T}v)}_u + \mathcal{T}v.$$

Now let’s calculate the slope of T in the direction of $\tilde{t} + cu$ as a function of c : how much

we perturb away from the direction of the steepest slope $\tilde{t}$. That is

$$s(c) = \frac{T(\tilde{t} + cu)}{\|\tilde{t} + cu\|} = \frac{T\tilde{t} + cTu}{\sqrt{1 + c^2\|u\|^2}}$$

where we've expanded the numerator using linearity and calculated the denominator with the pythagorean theorem ($u \perp \tilde{t}$ by construction). Since the slope is maximized in the direction $\tilde{t}$, we know that $s(c)$ has its maximum at $c = 0$, meaning that the derivative is zero there: $s'(0) = 0$. However, from the above, a tedious application of the product and chain rules and evaluating at $c = 0$ gives $s'(0) = Tu$. But since we already know that must be zero, we conclude $Tu = 0$. In other words, T does not care about any component of its input that is not aligned with the direction of maximum increase $\tilde{t}$.

As a consequence of this,

$$\begin{aligned} T(v) &= T(u + \mathcal{T}v) \\ &= T(\mathcal{T}v) \\ &= T(\langle \tilde{t}, v \rangle \tilde{t}) \\ &= \langle T(\tilde{t}), v \rangle \tilde{t} \end{aligned}$$

Where we've used the 1D projection formula $\mathcal{T}v = \langle \tilde{t}, v \rangle \tilde{t}$ and linearity to move terms around. This establishes that $T(\tilde{t}) \tilde{t}$ is the Riesz representer of T . The result once again aligns with the intuition from the finite-dimensional case: the representer is the (unit) direction of steepest ascent, scaled by the slope of the function in that direction.

Exercise B.11. Is $f \mapsto E[f(Z)]$ a bounded linear functional over all of $\mathcal{L}_2(\mathbb{P})$? If so, what is its Riesz representer?

Exercise B.12. For a bounded functional T , prove that $\|T\|_{\text{op}} = \|t\|$ where t is the Riesz representer of T .

If you are paying very, very close attention, you may have noticed that the pointwise evaluation functional is actually *not* well-defined on $\mathcal{L}_2(\mathbb{P})$. The reason is that, as you may recall, the elements of $\mathcal{L}_2(\mathbb{P})$ are not really functions, but *equivalence classes* of functions. For an operator that actually works on individual functions to be well-defined on $\mathcal{L}_2(\mathbb{P})$, it must have the same output for each function in any given equivalence class.

There *are* unbounded linear functionals that are well-defined on $\mathcal{L}_2$ but it's actually not possible to construct one explicitly. Proving what they look like *implicitly* requires more functional analysis than I'd like to get into in this book. Moreover, the examples of "unbounded linear functionals" that come up in statistics are more often like my technically ill-defined example: functionals that are actually directly defined on a space of individual functions, but we pretend are defined on $\mathcal{L}_2$. Ultimately it doesn't make much of a difference because the conclusion is

the same either way: Riesz representation doesn't work for these functionals.